When Risk Becomes Ambiguity: Shadow Banking, Collateral, and the Left Tail

Lei Pan[*†]

Curtin University and Monash University

Abstract

This paper develops a theoretical model of how financial uncertainty reshapes macroeconomic downside risk through shadow banking and collateral feedback. Financial uncertainty is modelled as ambiguity about firm fundamentals: when lenders become less confident about the probability law governing repayment, they behave as if their effective risk aversion rises. This widens credit spreads, contracts lending, and raises default thresholds. The effect is stronger for shadow banks because rollover-sensitive funding makes them more exposed to ambiguous survival probabilities. Once shadow-bank net worth is exhausted, fire-sale losses transmit through interbank claims to regulated banks, generating an endogenous contagion threshold. At the same time, falling credit depresses collateral prices, weakens pledgeable wealth, and further tightens borrowing constraints. The model also shows that innovative firms are disproportionately exposed because their technologies are more capital intensive, riskier, and less pledgeable in default. Quantitative exercises show that ambiguity shifts the output distribution left, lowers the fifth percentile of output, and makes optimal capital regulation state contingent: buffers should be built in calm periods and released under high ambiguity.

Keywords: Financial uncertainty; ambiguity; downside risk; shadow banking; collateral feedback; macroprudential policy

JEL classification: D81; E32; G01; G21; G28

1 INTRODUCTION

A financial crisis rarely begins with a single observable loss. More often, it begins with a change in the meaning of information. A loan that looked safe yesterday becomes difficult to price today; collateral that seemed liquid becomes conditional on the willingness of other investors to buy; a shadow-bank liability that was rolled over routinely becomes fragile once creditors ask what, exactly, stands behind it. In such moments the central problem is not simply that fundamentals deteriorate. It is that lenders no longer know how to attach probabilities to fundamentals. The economy moves from risk to ambiguity. Balance sheets that were designed for a world of measurable default probabilities are forced to operate in a world in which model

[*]Corresponding author. School of Accounting, Economics and Finance, Curtin University, Kent Street, Bentley, Perth WA 6102, Australia. Email: `lei.pan@curtin.edu.au`.

[†]Centre for Development Economics and Sustainability, Monash University, Clayton, VIC-3800, Australia.

confidence itself has declined. Credit spreads widen, lending contracts, collateral prices fall, and the losses that matter most are not average losses but left-tail losses: the outcomes that materialize when uncertainty, funding fragility, and collateral feedback reinforce one another.

This paper develops a theoretical framework for understanding how financial uncertainty moves the left tail of macroeconomic outcomes through shadow banking and collateral. The starting point is a simple observation. Modern credit systems are not organized around a single representative lender. Regulated banks and shadow banks coexist, share borrowers, and are linked through interbank claims, wholesale funding, and collateral values. When ambiguity about firm fundamentals rises, both types of intermediaries become more cautious, but they do not respond symmetrically. The shadow bank is more exposed to rollover risk and therefore prices ambiguity more aggressively. Its credit supply contracts more strongly. If the shock is large enough, the shadow bank's own net worth is exhausted, fire sales begin, and losses are transmitted back to the commercial bank through an interbank claim. At the same time, contracting credit lowers firms' demand for pledgeable assets, depresses the collateral price, and further reduces borrowing capacity. The result is an endogenous amplification mechanism: financial uncertainty first tightens credit directly, then deepens the downturn through network contagion and collateral feedback.

The mechanism is especially important for innovative firms. Innovation is often financed against assets that are difficult to pledge or redeploy: organizational capital, intangible capital, specialized knowledge, and frontier technologies. In the model, firms choose between a traditional technology and an innovative technology. The innovative technology is more capital-intensive, has higher frontier risk, and has lower recovery value in default. These features are not imposed as a reduced-form "type" effect. They arise from primitive differences in capital intensity, pledgeability, and idiosyncratic risk. As a result, an increase in ambiguity does more than reduce total credit. It reallocates activity away from innovation. The share of firms adopting the innovative technology falls, and the innovative sector suffers a larger deterioration in downside output. This mechanism connects financial uncertainty to the composition of real activity, not only to its aggregate level.

This paper speaks to several strands of the literature. First, it builds on the macroeconomic literature on uncertainty shocks, which shows that uncertainty can generate sharp declines in investment, employment, and output [Bloom, 2009, 2014], and on the measurement literature that constructs empirical proxies for policy, macroeconomic, and financial uncertainty [Baker et al., 2016, Jurado et al., 2015, Cascaldi-Garcia et al., 2023]. Related work has examined whether uncertainty is an exogenous impulse or an endogenous response to the business cycle [Ludvigson et al., 2021], how uncertainty interacts with financial shocks [Caldara et al., 2016], and how uncertainty affects aggregate demand and monetary transmission [Basu and Bundick, 2017, Aksoy and Basso, 2014]. Second, the paper connects to the financial-accelerator and credit-supply literatures, in which balance-sheet strength and external-finance premia amplify real fluctuations [Bernanke et al., 1999, Kiyotaki and Moore, 1997]. Empirical studies on bank lending show that monetary and liquidity shocks are transmitted through credit supply and bank balance sheets [Kashyap et al., 1993, Kashyap and Stein, 2000, Khwaja and Mian, 2008, Jiménez et al., 2012, 2014]. The Growth-at-Risk approach further emphasizes that financial conditions are especially informative about the lower tail of future growth rather than only about the conditional mean [Adrian et al., 2019]. Third, the paper relates to the shadow-banking literature, which highlights leverage, wholesale funding, repo runs, and global liquidity as sources of systemic fragility [Adrian and Shin, 2010, Pozsar et al., 2010, Adrian and Ashcraft, 2012, Gorton and Metrick, 2012, Bruno and Shin, 2015, Moreira and Savov, 2017]. Fourth, it contributes to the collateral and housing-macro literatures, where borrowing constraints, house prices, land values, and collateral values propagate shocks into investment and output [Iacoviello, 2005, Davis and Heathcote, 2005, Iacoviello and Neri, 2010, Liu et al., 2013, Chaney et al., 2012, Gan, 2007, Wu et al., 2015, Saiz, 2010]. Fifth, it is related to macroprudential-policy research on risk-taking, regulatory leakage, capital buffers, and the effectiveness of prudential tools across the cycle [Borio and Zhu, 2012, Dell'Ariccia et al., 2014, Aiyar et al., 2014, Cerutti et al., 2017, Akinci and Olmstead-Rumsey, 2018, Altunbas et al., 2018, Jiménez et al., 2017, Bianchi and Mendoza, 2018]. Sixth, the model is connected to the literature on innovation finance, which stresses that R&D and high-tech investment are particularly dependent on internal finance, external equity, and financial-market conditions [Himmelberg and Petersen, 1994, Carpenter and Petersen, 2002, Brown et al., 2009, Hall and Lerner, 2010]. Finally, the formal treatment of ambiguity is grounded in models of smooth ambiguity, robustness, and ambiguity in asset markets [Klibanoff et al., 2005, Hansen and Sargent, 2008, Epstein and Schneider, 2010], while the contagion and fire-sale channels are related to work on inefficient credit booms, financial networks, interbank contagion, and liquidation values [Lorenzoni, 2008, Acemoglu et al., 2015, Allen and Gale, 2000, Shleifer and Vishny, 1992, Acharya et al., 2007, Almeida and Campello, 2007, Kermani and Ma, 2023].

The paper makes four contributions. First, it gives financial uncertainty a decision-theoretic foundation. Instead of treating uncertainty as an exogenous spread shifter, the model derives an effective-risk-aversion channel from smooth ambiguity preferences. Lenders face a family of models over firm survival probabilities, and greater dispersion over these models raises their ambiguity-adjusted valuation of the low-payoff default state. This generates an endogenous mapping from ambiguity to credit spreads and credit supply. Importantly, the pass-through is steeper for shadow banks because rollover-sensitive funding increases their exposure to ambiguous survival probabilities. The model therefore explains why shadow credit is more uncertainty-elastic without assuming that shadow banks are mechanically more fragile.

Second, the paper develops a unified amplification mechanism that combines shadow-bank fragility, interbank contagion, and collateral feedback. Contagion is not imposed as an exogenous loss. It arises when shadow-bank net worth is exhausted and forced liquidation reduces the value of the interbank claim held by the commercial bank. The contagion threshold is therefore an equilibrium object. Collateral prices are also endogenous: when ambiguity tightens credit, firms demand fewer pledgeable assets, the collateral price falls, pledgeable wealth shrinks, and credit tightens further. This creates a feedback multiplier between the financial sector and the collateral market. The model thus shows how two amplifiers that are often studied separately, network spillovers and collateral-price feedback, interact to move the lower tail of output.

Third, the paper introduces an endogenous technology-adoption margin into a financial uncertainty model. Innovative firms are not assumed to be fragile simply by label. They are more exposed because their technology is more capital-intensive, riskier at the frontier, and less pledgeable in default. These primitives raise both the actuarial and ambiguity components of borrowing costs. Higher ambiguity therefore reduces the relative profitability of innovation, lowers the equilibrium share of innovative firms, and widens the downside-risk gap between innovative and traditional production. The model consequently links financial uncertainty to long-run allocative damage: a crisis can depress not only the level of output but also the economy's movement toward high-productivity technologies.

Fourth, the paper delivers a state-contingent macroprudential implication. Capital requirements improve resilience when ambiguity is low, but when ambiguity is high the dominant forces behind the left tail are ambiguity pass-through, shadow-bank contraction, and contagion. In that region, additional capital requirements can impose a large credit cost while doing little to remove the source of the tail shock. The model therefore predicts that the optimal capital ratio is decreasing in ambiguity: buffers should be built in calm times and released in stress. This result complements the empirical and theoretical literature on countercyclical macroprudential policy, while clarifying a specific mechanism through which tight regulation in a high-ambiguity state may deepen the left tail.

The model is organized as follows. Firms choose between a traditional and an innovative technology, choose investment scale, and borrow from either a regulated commercial bank or a shadow bank. Intermediaries observe the degree of ambiguity about firm fundamentals and set credit terms. The commercial bank is subject to a capital requirement and lends against housing collateral; the shadow bank is outside the capital requirement but is exposed to rollover risk. A household sector owns the fixed housing stock, and market clearing determines the equilibrium collateral price. Defaults occur after productivity is realized. If the shadow bank's net worth is exhausted, fire-sale losses reduce the value of the commercial bank's interbank claim. The macroprudential authority chooses the capital requirement and the loan-to-value cap to maximize the fifth percentile of output, $Y_{0.05}$, subject to maintaining sufficient credit supply.

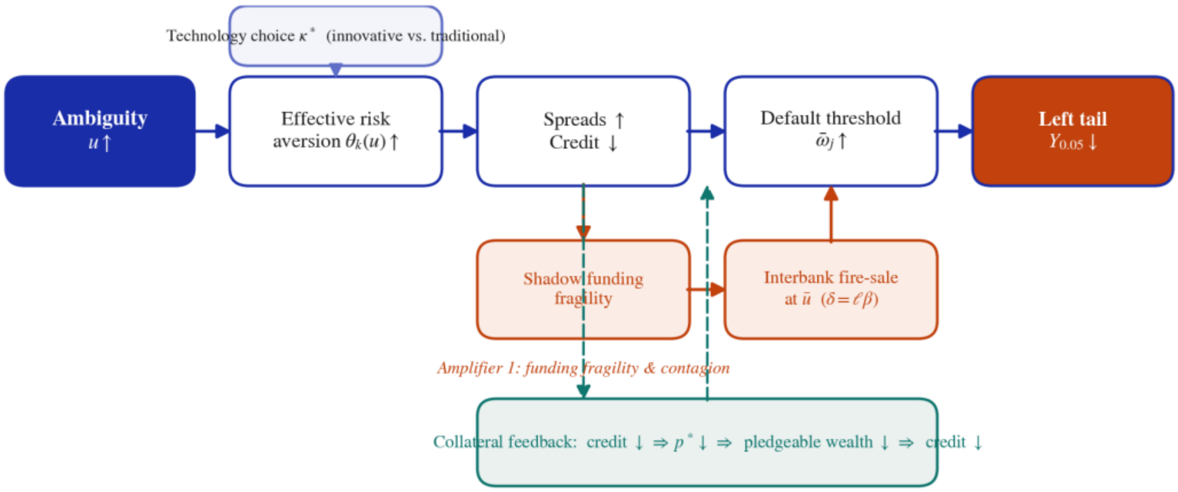
The main results are transparent. Ambiguity raises effective risk aversion, increases credit spreads, and contracts lending by both intermediaries. Shadow credit contracts more because the shadow bank's funding structure makes its continuation value more sensitive to ambiguous survival probabilities. Higher spreads raise default thresholds and increase fragility. Once ambiguity crosses the endogenous shadow-bank solvency threshold, interbank contagion switches on and total credit contracts more sharply than in a no-contagion benchmark. Endogenous collateral prices magnify the effect: weaker housing fundamentals make the ambiguity–downside-risk relationship more severe. Innovative firms bear the largest tail losses because their assets recover less in default, their technology is riskier, and their output is more sensitive to investment. Finally, the optimal capital requirement declines with ambiguity, reflecting the changing balance between resilience benefits and credit-supply costs.

Numerical illustrations support the analytical mechanism. As ambiguity rises, credit spreads increase, direct credit supply falls, and the effective risk aversion of the shadow bank rises more steeply than that of the commercial bank. The output distribution shifts left, with the fifth percentile declining substantially more than the mean. Counterfactual exercises show that turning off interbank contagion or freezing the collateral price raises the lower tail relative to the full model, confirming that both channels deepen downside risk. Heterogeneity exercises show that the innovative sector experiences a larger deterioration in $Y_{0.05}$, while policy experiments show that the capital ratio that maximizes downside output shifts downward as ambiguity increases. These results suggest a central lesson: when the economy is governed by ambiguity rather than measurable risk, macroprudential policy must be designed for the tail, not only for average credit conditions.

2 THE MODEL

The economy lasts one period with an interim stage. There are four sectors: a unit mass of firms that choose a production technology and borrow; two intermediaries (a regulated commercial bank B and a shadow bank S) linked by an interbank claim; and a household sector that owns the housing stock and supplies deposits. Figure 1 summarizes the timing. At T_0 firms choose technology, scale, and lender. At T_1 intermediaries observe the degree of ambiguity u about firm fundamentals and set credit terms; the housing market clears. At T_2 productivity is realized, defaults occur, and, if shadow-bank net worth is exhausted, interbank losses propagate. At T_3 output and payoffs are settled.

Figure 1. Timeline with technology choice, ambiguity, and interbank linkage



Notes: At T_0 firms choose between the tangible and intangible technologies and a lender. At T_1 intermediaries observe ambiguity u and price credit; the household–firm housing market clears at price $p^*(u)$. At T_2 productivity realizes and defaults occur; if shadow-bank net worth is exhausted, fire-sale losses hit commercial banks' interbank claims. At T_3 payoffs settle.

2.1 Firms and the technology-adoption margin

Rather than defining firm types by their output distributions, we let heterogeneity arise from an adoption decision over a technology menu. This makes the riskier profile of innovative firms a consequence of a pledgeability trade-off, and makes the composition of the firm population an equilibrium object that responds to uncertainty.

Assumption 1 (Technology menu)**.** *A firm can operate one of two technologies, $j \in \{L, H\}$, each with production $y_j = \omega_j I_j^{\alpha_j}$ and $\ln \omega_j \sim \mathcal{N}(0, \sigma_j^2)$. The technologies differ in three primitives:*

(i) Capital intensity: $\alpha_H > \alpha_L$ (innovation is more capital-intensive);

(ii) Pledgeability: upon default a lender recovers a fraction $r^j_{\text{rec}} \in (0,1)$ of the loan, with $r^H_{\text{rec}} < r^L_{\text{rec}}$ because intangible capital is harder to redeploy;

(iii) Frontier risk: operating at the technological frontier carries more idiosyncratic dispersion, $\sigma_H > \sigma_L$.

Adopting H requires an idiosyncratic sunk cost $c \sim G(\cdot)$ on $[0, \infty)$ (organizational capital, talent, tooling), drawn independently across firms.

Here $\sigma_H > \sigma_L$ and $\alpha_H > \alpha_L$ are properties of the intangible technology, and lower pledgeability r^H_{rec} is a third, independent primitive; none is the definition of "type." A firm adopts H when the expected-profit gain exceeds its draw c, so the equilibrium share of innovators is

$$\kappa^* = G\Big(\Pi_H - \Pi_L\Big), \qquad \Pi_j \equiv \max_{I_j} \; \mathbb{E}\Big[\pi_j(I_j)\Big], \tag{1}$$

where π_j is firm profit defined below. Because $\Pi_H - \Pi_L$ depends on credit terms, κ^* is endogenous and, as shown in Section 3, falls with uncertainty.

A firm that has chosen technology j and borrowed I_j at all-in cost $R + \Delta R_k$ defaults if productivity is too low. The break-even (default) threshold solves the zero-net-revenue condition

$$\bar{\omega}_j = I_j^{1-\alpha_j}\Big(R + \Delta R_k\Big) + I_j^{1-\alpha_j} M, \tag{2}$$

where $R > 1$ is the gross risk-free rate, $\Delta R_k \geq 0$ the spread charged by lender $k \in \{B, S\}$, and $M > 0$ a per-unit liquidation cost. The firm chooses scale to maximize expected profit,

$$\max_{I_j} \quad I_j^{\alpha_j} \int_{\bar{\omega}_j}^{\infty} \omega \, dF_j(\omega) - \Big[1 - F_j(\bar{\omega}_j)\Big]\Big[I_j(R + \Delta R_k) + M\Big], \tag{3}$$

with first-order condition (F.O.C)

$$\alpha_j I_j^{\alpha_j - 1} \int_{\bar{\omega}_j}^{\infty} \omega \, dF_j(\omega) - \Big[1 - F_j(\bar{\omega}_j)\Big](R + \Delta R_k) = 0, \tag{4}$$

and the marginal zero-profit condition reproduces Equation (2):

$$\bar{\omega}_j = I_j^{1-\alpha_j}(R + \Delta R_k) + I_j^{1-\alpha_j} M. \tag{5}$$

Realized output mixes defaulting and surviving firms,

$$Y_j = \begin{cases} (1-\gamma)\, \omega_j I_j^{\alpha_j}, & \omega_j < \bar{\omega}_j \quad \text{(default)}, \\ \omega_j I_j^{\alpha_j}, & \omega_j \geq \bar{\omega}_j \quad \text{(survival)}, \end{cases} \tag{6}$$

with $\gamma \in (0, 1)$ the output loss in default. We track the 5th percentile $Y_{0.05}$ of the output distribution as the measure of downside (Growth-at-Risk) exposure.

2.2 Financial uncertainty as ambiguity

We now give "financial uncertainty" a decision-theoretic content. Intermediaries do not know the true productivity law F_j; they entertain a family of models indexed by m, with model-implied survival probability $s(m) \equiv 1 - F_j(\bar{\omega}_j \mid m)$. A lender holds a second-order belief μ over m with mean $\bar{s}$ and variance

$$\mathrm{Var}_{\mu}\Big(s(m)\Big) = u^2 \varsigma^2, \tag{7}$$

so $u \geq 0$ scales the dispersion of beliefs about fundamentals: $u = 0$ is risk (a known F_j); $u > 0$ is Knightian ambiguity. Preferences are of the smooth-ambiguity form [Klibanoff et al., 2005],

$$\mathcal{V} = \mathbb{E}_{\mu}\Big[v\Big(\mathbb{E}_m[U(W)]\Big)\Big], \qquad v(x) = -\tfrac{1}{\eta} e^{-\eta x}, \quad \eta > 0, \tag{8}$$

where U is the lender's (CRRA) utility over terminal wealth W, $\mathbb{E}_m$ is the expectation under model m, and η is the coefficient of ambiguity aversion. A second-order expansion of (8) around the mean model yields the following observational-equivalence result, proved in Appendix A.

Lemma 1 (Ambiguity as effective risk aversion)**.** *To second order in u, the smooth-ambiguity lender in (8) behaves as an expected-utility lender with effective risk aversion*

$$\theta_k(u) = \theta_k^0 + \phi_k \, u, \qquad \phi_k = \Lambda\Big(\eta, \, \mathcal{E}_k\Big) > 0, \tag{9}$$

where θ_k^0 is the lender's intrinsic risk aversion, and the slope ϕ_k is increasing in ambiguity aversion η and in the balance-sheet exposure $\mathcal{E}_k$ of the lender's certainty-equivalent to the ambiguous survival rate $s(m)$. Hence $\partial\theta_k/\partial u = \phi_k > 0$ is derived, not assumed, and the transmission slope

ϕ_k inherits the curvature of preferences and the lender's exposure.

Proof. *See Appendix A.* □

Equation (9) is the same object used informally in earlier reduced-form models, but here it is the conclusion of an optimization under ambiguity: u is genuine uncertainty about technology, and the pass-through ϕ_k is a function of primitives. This is what allows the comparative statics in u to be read as the effects of uncertainty rather than of an unexplained taste shock. The lender's CRRA felicity is $U_k(W) = W^{1-\theta_k}/(1-\theta_k)$.

2.3 Commercial bank

The commercial bank has equity N_B and lends b_B. It is repaid $b_B(R + \Delta R_B)$ if the firm survives and recovers $r^j_{\text{rec}} b_B$ in default. It faces a macroprudential capital requirement $\tau \in (0, 1)$. Using Lemma 1, the bank maximizes effective expected utility

$$\mathbb{E}[U_B] = F_j(\bar{\omega}_j)\frac{(N_B + r^j_{\text{rec}} b_B - R b_B)^{1-\theta_B}}{1-\theta_B} + \Big[1 - F_j(\bar{\omega}_j)\Big]\frac{(N_B + b_B \Delta R_B)^{1-\theta_B}}{1-\theta_B}. \tag{10}$$

With housing collateral (subsection 2.6) the survival-state payoff is augmented by the pledged value $\lambda p \chi$. Optimization subject to the binding capital constraint yields the closed-form supply

$$b^*_B = \frac{N_B(1+\tau) + \lambda p \chi}{\tau + \theta_B(u)\, u\, \xi}, \tag{11}$$

where $\xi > 0$ collects the curvature terms from the first-order condition (full derivation in Appendix A, which records the auxiliary objects $W^{\text{def}}_B, W^{\text{norm}}_B$ and the FOC). Equation (11) makes transparent that credit falls when the derived effective risk aversion $\theta_B(u)$ rises and increases in pledgeable collateral $\lambda p \chi$.

2.4 Shadow bank and funding fragility

The shadow bank has equity N_S and lends b_S, funding a fraction of its assets with short-term wholesale debt that must be rolled at T_2. It is outside the capital requirement but its continuation value is more exposed to the ambiguous survival rate, because rollover failure is triggered precisely in low-$s(m)$ models. Formally, its certainty-equivalent exposure satisfies $\mathcal{E}_S > \mathcal{E}_B$, so Lemma 1 delivers a steeper pass-through.

Lemma 2 (Funding fragility steepens pass-through)**.** *If the shadow bank funds a positive share of assets with rollover-sensitive debt while the commercial bank funds with insured deposits, then $\mathcal{E}_S > \mathcal{E}_B$ and therefore*

$$\theta_S(u) = \theta^0_S + \phi_S u, \qquad \phi_S > \phi_B. \tag{12}$$

The inequality $\phi_S > \phi_B$ is thus an implication of funding structure, not a primitive ranking.

Proof. *See Appendix A.* □

Absent the capital requirement and collateral facility, the shadow bank's supply is

$$b^*_S = \frac{N_S}{1 + \theta_S(u)\, u\, \zeta}, \qquad \zeta > 0. \tag{13}$$

2.5 Interbank market and endogenous contagion

We now derive contagion from balance-sheet linkages and fire-sale pricing rather than imposing it. A fraction $\beta \in (0, 1)$ of the shadow bank's funding is an interbank claim held by the commercial bank, $L_{\text{int}} = \beta\, b_S$. At T_2 the shadow bank's net worth is

$$n_S(u) = N_S - \Big[1 - s(\cdot)\Big]\, \Theta(u)\, b_S, \tag{14}$$

where $\Theta(u)$ is the (ambiguity-inflated) expected loss rate on shadow-bank assets, increasing in u. Define the contagion threshold $\bar{u}$ as the ambiguity level at which shadow-bank net worth is exhausted,

$$n_S(\bar{u}) = 0 \iff \bar{u} : \Big[1 - s(\cdot)\Big]\, \Theta(\bar{u})\, b_S = N_S. \tag{15}$$

For $u > \bar{u}$ the shadow bank must liquidate assets into an illiquid market at a fire-sale price $q(u) = 1 - \ell\,(u - \bar{u})$, with $\ell > 0$ the market-illiquidity (price-impact) parameter [Lorenzoni, 2008, Acemoglu et al., 2015]. The commercial bank's interbank claim is then written down by $(1 - q(u))L_{\text{int}}$, so its effective capital becomes

$$\widetilde{N}_B(u) = N_B - \underbrace{\ell\,\beta}_{\equiv\, \delta}\, \Big(u - \bar{u}\Big)\, b_S\, \mathbf{1}\{u > \bar{u}\}, \tag{16}$$

which is exactly the contagion term used below, but now $\delta = \ell\beta$ is a product of primitives (price impact $\times$ interbank exposure) and the threshold $\bar{u}$ is the endogenous solvency point in Equation (15). The earlier "assumption" of contagion is therefore a *lemma*: it is the accounting consequence of an interbank claim revalued at a fire-sale price.

2.6 Households and the housing market

Collateral prices are not exogenous. A unit mass of households owns the fixed housing stock $\bar{H}$, supplies deposits to the commercial bank, and has downward-sloping housing demand $H^h(p) = a - b_h p$ with $a, b_h > 0$. Firms demand housing as a pledgeable asset; aggregate firm demand rises with investment (more activity requires more collateral capacity) and falls with its price, $H^f(p,u) = \gamma_f\, \mathcal{I}(p,u)$, where $\mathcal{I}(p,u) \equiv \sum_j (b_B^{j*} + b_S^{j*})$ is total credit-financed investment, increasing in p through Equation (11) and decreasing in u. Market clearing $H^h(p) + H^f(p,u) = \bar{H}$ pins down the equilibrium collateral price.

Lemma 3 (Endogenous collateral price and feedback)**.** *Let $\mathcal{I}_p \equiv \partial\mathcal{I}/\partial p > 0$ and $\mathcal{I}_u \equiv \partial\mathcal{I}/\partial u < 0$, and assume the stability condition $b_h - \gamma_f \mathcal{I}_p > 0$. Then the market-clearing price is*

$$p^*(u) = \frac{a + \gamma_f\, \mathcal{I}(\,\cdot\,,u)\Big|_{p=0} - \bar{H}}{b_h - \gamma_f \mathcal{I}_p}, \qquad \frac{dp^*}{du} = \frac{\gamma_f\, \mathcal{I}_u}{b_h - \gamma_f \mathcal{I}_p} < 0. \tag{17}$$

The denominator $b_h - \gamma_f \mathcal{I}_p$ is the inverse of a collateral-feedback multiplier: a fall in p lowers pledgeable wealth, contracts credit and firm housing demand, and depresses p further. Comparative statics are taken with respect to fundamentals (housing supply $\bar{H}$, demand a, b_h, ambiguity u); the collateral price p^* is an equilibrium response, not an instrument.

Proof. *See Appendix A.* □

Substituting $p = p^*(u)$ into Equation (11) closes the credit$\leftrightarrow$collateral loop. We retain the convenient reduced form for the bank's collateralized supply,

$$b_B^* = \frac{N_B(1+\tau) + \lambda\, p^*(u)\, \chi}{\tau + \theta_B(u)\, u\, \xi}, \tag{18}$$

now understanding $p^*(u)$ as endogenous.

2.7 Credit market, spreads, and macroprudential policy

Total credit to type-j firms is $I_j = b_B^j + b_S^j$, and market clearing requires

$$I_j^* = b_B^{j*} + b_S^{j*}, \qquad j \in \{H, L\}. \tag{19}$$

The equilibrium spread charged by lender k has an actuarial component and an ambiguity (risk-aversion) component:

$$\Delta R_k^j = \frac{F_j(\bar{\omega}_j)\,(R - r_{\mathrm{rec}}^j)}{1 - F_j(\bar{\omega}_j)} + \psi_k\,\theta_k(u)\,\sigma_j, \tag{20}$$

with $\psi_k > 0$. Note the actuarial term now depends on the technology-specific recovery r_{rec}^j, so innovative firms pay more both because $\sigma_H > \sigma_L$ and because $r_{\mathrm{rec}}^H < r_{\mathrm{rec}}^L$: two distinct primitives. The collateral value entering bank supply is

$$C_j = \lambda\,p^*(u)\,\chi. \tag{21}$$

The macroprudential authority chooses the capital ratio τ and the LTV cap λ to maximize the economy's downside,

$$\max_{\tau,\lambda} \quad Y_{0.05}\Big(\tau, \lambda \mid u\Big) \qquad \text{s.t.} \qquad \sum_j \Big(b_B^{j*} + b_S^{j*}\Big) \;\geq\; \underline{I}. \tag{22}$$

Definition 1 (Equilibrium)**.** Given u and policy (τ, λ), an equilibrium is a technology-adoption share κ^*, firm scales $\{I_j^*\}$, intermediary supplies $\{b_B^{j*}, b_S^{j*}\}$, spreads $\{\Delta R_k^{j*}\}$, a contagion threshold $\bar{u}$, and a collateral price p^* such that: (i) firms optimize (3) and adopt per Equation (1); (ii) intermediaries optimize under effective risk aversion Equations (9)–(12); (iii) $\bar{u}$ solves Equation (15) and contagion follows Equation (16); (iv) the housing market clears, (17); and (v) the credit market clears, Equation (19).

3 RESULTS

We state the results as propositions, each preceded by the economic mechanism in words. All proofs are in Appendix A. Throughout, the objects $\theta_k(u)$, δ, $\bar{u}$, $p^*(u)$, and κ^* are the derived equilibrium quantities of Section 2, so the comparative statics below are statements about the primitives (ambiguity, funding structure, interbank exposure, pledgeability), not about free parameters.

3.1 Credit contracts, and shadow credit contracts more

Mechanism. Higher ambiguity raises effective risk aversion (Lemma 1). A more risk-averse lender values the low-payoff default state more heavily, demands a wider spread, and lends less. The shadow bank's continuation value is more exposed to the ambiguous survival rate (Lemma 2), so its effective risk aversion rises faster and its credit contracts more. The regulated bank is partly shielded by its capital buffer, which acts as an additive floor in its supply schedule.

Proposition 1 (Credit contraction)**.** *$\partial b_B^*/\partial u < 0$ and $\partial b_S^*/\partial u < 0$, and shadow credit is more uncertainty-elastic: $\Big|\partial b_S^*/\partial u\Big| > \Big|\partial b_B^*/\partial u\Big|$ in elasticity terms.*

Proof. *See Appendix A.* □

Corollary 1 (Accelerating total contraction). *Total credit $I_j^* = b_B^{j*} + b_S^{j*}$ satisfies $\partial I_j^*/\partial u < 0$ and $\partial^2 I_j^*/\partial u^2 < 0$: the contraction accelerates with ambiguity.*

Proof. *See Appendix A.* □

3.2 Default thresholds rise

Mechanism. Two forces push the break-even productivity $\bar{\omega}_j$ up. Spreads widen one-for-one with the repayment burden (a unit elasticity), while the offsetting fall in scale enters only with elasticity $1 - \alpha_j < 1$. The spread channel therefore dominates, and because spreads rise without bound as ambiguity grows while scale is bounded below, the threshold, and hence the default probability, rises.

Proposition 2 (Uncertainty and default). *For both technologies, $d\bar{\omega}_j^*/du > 0$ and $dF_j(\bar{\omega}_j^*)/du > 0$.*

Proof. *See Appendix A.* □

Corollary 2 (Convex fragility). *When $\bar{\omega}_j^*$ lies below the mode of f_j, $d^2 F_j(\bar{\omega}_j^*)/du^2 > 0$: marginal fragility rises with ambiguity.*

Proof. *See Appendix A.* □

3.3 Contagion is an endogenous, threshold amplifier

Mechanism. Below the solvency point $\bar{u}$ in Equation (15), the shadow bank absorbs losses in its own equity and the interbank claim is money-good. Once u exhausts shadow-bank net worth, forced liquidation depresses the fire-sale price, the commercial bank's interbank claim is written down, its effective capital falls, and its lending contracts beyond the direct ambiguity effect. The extra term is proportional to the price impact ℓ and the interbank exposure β, and switches on only at $\bar{u}$—so the amplifier and its trigger are both equilibrium objects.

Proposition 3 (Network-contagion amplification). *For $u > \bar{u}$,*

$$\frac{d(b_B^* + b_S^*)}{du} = \underbrace{\frac{\partial b_B^*}{\partial u}}_{\text{(I) direct}} + \underbrace{\frac{db_S^*}{du}}_{\text{(II)}} + \underbrace{\frac{\partial b_B^*}{\partial \tilde{N}_B}\frac{d\tilde{N}_B}{du}}_{\text{(III) contagion, } <0},$$

so total credit contracts strictly more than in the no-contagion benchmark ($\delta = \ell\beta = 0$), and $d^2(b_B^* + b_S^*)/du^2$ jumps down at $u = \bar{u}$.

Proof. *See Appendix A.* □

Corollary 3 (Contagion multiplier). *The amplification factor $\mathcal{M}(u) = 1 + |\text{(III)}|/(|\text{(I)}| + |\text{(II)}|) > 1$ for $u > \bar{u}$ is increasing in price impact ℓ, interbank exposure β, and the shadow bank's market share.*

Proof. *See Appendix A.* □

3.4 Collateral feedback magnifies downside risk

Mechanism. The collateral price is now endogenous (Lemma 3). Higher ambiguity tightens credit, which lowers firms' demand for pledgeable assets and depresses the equilibrium house price; the lower price shrinks collateral and tightens credit again. The strength of this loop is the collateral-feedback multiplier $1/(b_h - \gamma_f \mathcal{I}_p)$. Because output is concave in investment, a given proportional credit loss costs more output when the economy starts from a depressed (low-collateral) state, so weak fundamentals and high ambiguity are complements for downside risk.

Proposition 4 (Collateral-feedback amplification)**.** *$dp^*/du < 0$, and the effect of ambiguity on downside risk is larger (more negative) the weaker are housing fundamentals:*

$$\frac{\partial^2 Y_{0.05}}{\partial u\, \partial \bar{H}} > 0, \qquad \textit{equivalently} \qquad \frac{\partial^2 Y_{0.05}}{\partial u\, \partial p^*} > 0.$$

The amplification is increasing in the feedback multiplier $1/(b_h - \gamma_f \mathcal{I}_p)$.

Proof. *See Appendix A.* □

Corollary 4 (LTV interaction)**.** *$\partial^2 Y_{0.05}/(\partial u\, \partial \lambda) > 0$: a tighter LTV cap amplifies the ambiguity–downside-risk transmission, the policy tension formalized in Proposition 6.*

Proof. *See Appendix A.* □

3.5 Innovative firms are more exposed—and the margin shifts

Mechanism. The result is not definitional. Innovative firms are more exposed for three reasons rooted in technology primitives: their collateral recovers less ($r^H_{\text{rec}} < r^L_{\text{rec}}$), which raises the actuarial part of their spread independently of dispersion; their frontier risk σ_H raises the ambiguity part of the spread; and their capital intensity α_H amplifies the output cost of any credit loss. Moreover, the share of innovators κ^* is endogenous and falls with ambiguity, so uncertainty also reallocates activity away from the innovative sector.

Proposition 5 (Differential firm vulnerability)**.** *$\left|\partial Y^H_{0.05}/\partial u\right| > \left|\partial Y^L_{0.05}/\partial u\right|$, driven by the recovery gap $r^L_{\text{rec}} - r^H_{\text{rec}} > 0$, the dispersion gap $\sigma_H - \sigma_L > 0$, and the capital-intensity gap $\alpha_H - \alpha_L > 0$. The adoption share satisfies $d\kappa^*/du < 0$.*

Proof. *See Appendix A.* □

Corollary 5 (Divergence)**.** *The vulnerability gap widens with ambiguity: $\partial\Big(|\partial Y^H_{0.05}/\partial u| - |\partial Y^L_{0.05}/\partial u|\Big)/\partial u > 0$.*

Proof. *See Appendix A.* □

3.6 State-contingent macroprudential policy

Mechanism. Capital requirements trade stability against credit. When ambiguity is low, the binding concern is leverage and the buffer mainly improves resilience; when ambiguity is high, the dominant drivers of the left tail are the ambiguity pass-through and contagion, which the buffer does not directly offset, while its credit cost is most acute. Hence the marginal benefit of tightening falls with ambiguity and the optimal buffer is countercyclical.

Proposition 6 (Macroprudential trade-off)**.** *There is an interior optimal capital ratio $\tau^*(u)$ maximizing $Y_{0.05}$, and it is decreasing in ambiguity: $d\tau^*/du < 0$.*

Proof. *See Appendix A.* □

Remark 1. The result rationalizes countercyclical capital buffers [Jiménez et al., 2017, Bianchi and Mendoza, 2018]: relax in stress, rebuild in calm. The caveat is moral hazard from anticipated forbearance, which is outside the one-period model.

4 NUMERICAL ILLUSTRATION

The propositions are qualitative. We now calibrate the model both to attach magnitudes to the mechanisms and to verify numerically that the derived equilibrium objects, the effective risk aversions $\theta_k(u)$, the endogenous contagion threshold $\bar{u}$, the market-clearing collateral price $p^*(u)$, and the innovation share κ^*, move in the directions the analysis predicts. This section interprets the calibration (Table 1) and the two baseline figures; Section 5 then turns off individual channels.

4.1 Calibration

Table 1 collects the baseline parameters. They fall into five blocks. What matters for the results is not the absolute levels of the balance-sheet quantities-, these are normalizations, expressed in units in which $N_B = 0.30$, but the inequalities the microfoundations of Section 2 require, and, for the new blocks, anchoring each parameter to an empirical benchmark. The technology block encodes the two primitives that separate innovative from traditional firms: innovative production is more capital-intensive ($\alpha_H = 0.55 > \alpha_L = 0.45$) nd operates with more frontier risk ($\sigma_H = 0.6 > \sigma_L = 0.4$), and, crucially for the heterogeneity result, its collateral is less pledgeable, with a recovery rate of $r^H_{\text{rec}} = 0.3$ against $r^L_{\text{rec}} = 0.6$. These recovery values are disciplined by the evidence that tangible, redeployable assets recover a large fraction of face value at default while intangible, industry-specific assets recover far less: senior secured claims on tangible assets recover on the order of 60–80%, whereas estimates for specific or intangible capital range from roughly 10% to 50% [Shleifer and Vishny, 1992, Acharya et al., 2007, Almeida and Campello, 2007, Kermani and Ma, 2023]. The recovery gap is an independent channel, so the model does not rely on the dispersion gap alone. The intermediary block sets a smaller shadow-bank balance sheet ($N_S < N_B$) and a lower intrinsic but more uncertainty-sensitive shadow risk attitude ($\theta^0_S = 1.5 < \theta^0_B = 2.0$ but $\phi_S = 2.5 > \phi_B = 1.5$); the inequality $\phi_S > \phi_B$ is not a free choice but the calibrated counterpart of Lemma 2, with ϕ_k interpreted through $\phi_k = \Lambda(\eta, \mathcal{E}_k)$ in Lemma 1. The ambiguity premium loads on credit terms through $\psi_B = 0.30 < \psi_S = 0.37$, and the supply schedules share a common curvature $\xi = \zeta = 0.06$. The linkage block sets the composite contagion parameter $\delta = \ell\beta = 0.40$ as a product of a stressed fire-sale price impact ($\ell = 0.8$) and a sizable interbank-exposure share ($\beta = 0.5$), rather than a stand-alone elasticity; the funding-fragility intensity pins the solvency root at $\bar{u} = 0.30$. The policy and collateral block uses Basel-style values ($\tau = 0.08$, $\lambda = 0.7$), a housing endowment $\chi = 0.2$, and a household–housing market whose supply elasticity is set to $\varepsilon = 1.75$, the population-weighted average of the metropolitan estimates in Saiz [2010]; the scarce- and ample-supply variants used below ($\varepsilon = 0.6$ and $\varepsilon = 3.0$) bracket his cross-sectional range. The default output loss $\gamma = 0.2$ sits inside the commonly cited 0.15–0.30 range. We emphasize that this is a data-anchored calibration, parameters set to empirically reasonable values from the literature, rather than a structural estimation; the goal is to attach plausible magnitudes to the mechanisms, and a full estimation on bank–firm and housing micro-data is left for future work.[1]

Table 1. Baseline calibration

Parameter	Symbol	Value	Rationale
Capital intensity (innovative)	α_H	0.55	Higher for intangible-intensive production
Capital intensity (traditional)	α_L	0.45	Standard
Risk-free gross rate	R	1.015	One-year benchmark
Frontier risk (innovative)	σ_H	0.6	Property of frontier technology
Frontier risk (traditional)	σ_L	0.4	Standard
Recovery, innovative / traditional	$r^H_{\text{rec}}, r^L_{\text{rec}}$	0.3 / 0.6	Intangibles redeploy poorly [Acharya et al., 2007, Kermani and Ma, 2023]
Liquidation cost	M	0.01	Small relative to scale
Output loss in default	γ	0.2	Empirical range 0.15–0.30
Bank / shadow equity	N_B, N_S	0.30 / 0.20	Normalized units; shadow smaller
Intrinsic risk aversion	θ^0_B, θ^0_S	2.0 / 1.5	CRRA
Ambiguity pass-through	ϕ_B, ϕ_S	1.5 / 2.5	$\phi_S > \phi_B$ from Lemma 2
Spread loadings	ψ_B, ψ_S	0.30 / 0.37	Shadow prices ambiguity more
Supply curvature	$\xi = \zeta$	0.06	Common normalization
Interbank exposure × price impact	$\delta = \ell\beta$	0.40	Stressed interbank (ℓ=0.8, β=0.5)
Contagion threshold (endogenous)	$\bar{u}$	0.30	Shadow-bank solvency root
LTV cap / capital ratio	λ, τ	0.7 / 0.08	Basel-style
Housing endowment	χ	0.2	Share of assets
Housing supply elasticity	ε	1.75	Population-weighted avg. [Saiz, 2010]
Housing demand (intercept, slope)	a, b_h	1.30 / 1.10	Household downward demand
Firm collateral demand	γ_f	0.35	Pledgeable-asset demand

The solution algorithm follows the equilibrium of Definition 1. For each ambiguity level $u \in [0.05, 1]$ we (i) evaluate the effective risk aversions $\theta_k(u) = \theta^0_k + \phi_k u$; (ii) solve the housing market for the clearing collateral price $p^*(u)$, recognizing that bank credit depends on p through pledgeable wealth and housing demand depends on credit, so p^* is a genuine fixed point; (iii) compute spreads ΔR^j_k and supplies b^{j*}_k (imposing the bank capital constraint), determine the technology-adoption share $\kappa^*(u)$ from the innovative–traditional profit gap, and solve the break-even thresholds $\bar{\omega}_j$; (iv) locate the contagion threshold $\bar{u}$ from the shadow-bank solvency condition (15) and apply the write-down (16) whenever $u > \bar{u}$; and (v) draw 80,000 productivity shocks across the type mixture to form the output distribution and read off its fifth percentile $Y_{0.05}$. The solvency condition delivers $\bar{u} \approx 0.30$, the market clears at $p^* \approx 0.78$ at low ambiguity, and the adoption share is interior throughout, so the figures below are generated by the derived equilibrium objects, not by the reduced-form expressions.

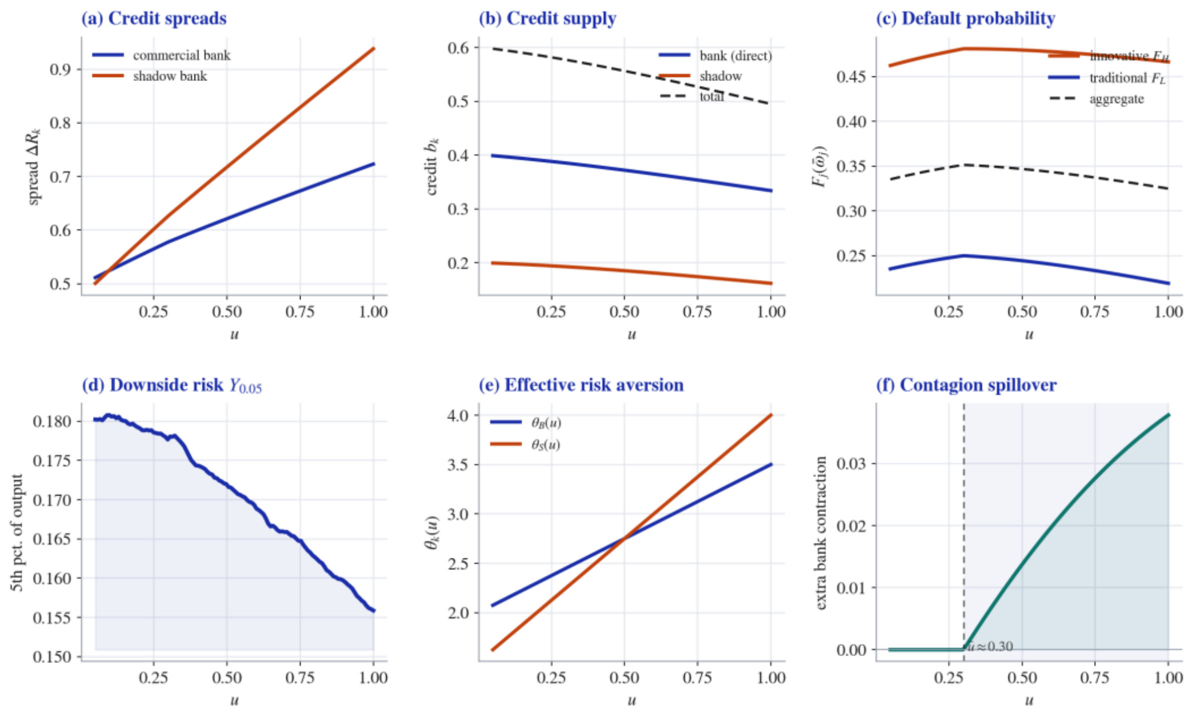
4.2 Equilibrium objects as ambiguity rises

Figure 2 traces six equilibrium objects against u and is best read as a visual proof of Propositions 1–3. Panel (a) plots credit spreads: both rise with ambiguity, and the shadow-bank schedule is steeper and fans out from the commercial-bank schedule as u grows—the two cross at low ambiguity and the shadow spread is roughly 0.22 above the bank spread by $u = 1$, the graphical signature of $\phi_S > \phi_B$ and of the widening spread differential in Equation (20). Panel (b) plots the direct credit-supply schedules: both contract, and shadow credit contracts faster in proportional terms (-19% against -16% for the bank over the range), so the shadow share of intermediation shrinks exactly as the system becomes more stressed (Proposition 1); total credit

[1] For numerical tractability the simulation evaluates the credit spread with the bounded expected-loss form $\Delta R^j_k = F_j(\bar{\omega}_j)\,(1 - r^j_{\text{rec}}) + \psi_k \theta_k(u)\sigma_j$ and the supply denominators with a unit normalization, $1 + \tau + \theta_B(u)u\xi$ and $1 + \theta_S(u)u\zeta$. Both are monotone transformations of the closed forms in Section 2 that keep credit finite as $u \to 0$ and prevent the actuarial term from diverging at high default; they leave every comparative static intact.

falls by about a quarter. Panel (c) shows default probabilities: the innovative threshold F_H sits far above the traditional F_L at every u (about 0.47 versus 0.23), and aggregate default is elevated and broadly flat-to-rising over the range as the higher spreads push up the break-even threshold $\bar{\omega}_j$ (Proposition 2)—the credit-contraction and spread channels nearly offset on the frequency of default, which is why the action is in the tail rather than the mean. Panel (d) reports the downside measure $Y_{0.05}$ falling monotonically, by about 14% from low to high ambiguity, confirming that ambiguity moves the tail. Panel (e) displays the derived effective risk aversions $\theta_B(u)$ and $\theta_S(u)$ rising linearly and crossing near $u = 0.5$, the calibrated image of Lemma 1; the steeper shadow line is the source of the asymmetry in panels (a)–(b). Panel (f) isolates the contagion spillover: the extra bank-credit contraction attributable to the interbank write-down: it is identically zero for $u < \bar{u}$ and switches on at $\bar{u} \approx 0.30$, after which it grows. The kink in panel (f) is the model's systemic-risk signature: nothing happens through the interbank channel until shadow-bank net worth is exhausted, and then losses propagate (Proposition 3).

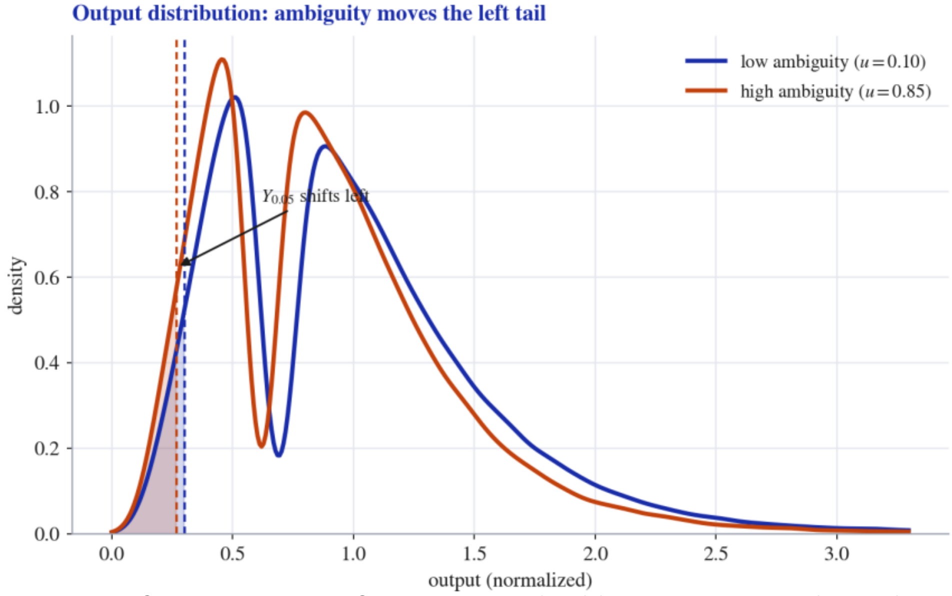
Figure 2. Equilibrium objects as functions of ambiguity



Notes: All objects computed under Table 1. Panel (a) credit spreads; (b) direct credit supplies (bank, shadow, total); (c) default probability by type and aggregate; (d) downside measure $Y_{0.05}$; (e) effective risk aversions $\theta_k(u)$; (f) the contagion spillover (extra bank-credit contraction), zero below the endogenous threshold $\bar{u} \approx 0.30$ and growing above it.

Figure 3 makes the tail effect concrete by overlaying the full output distribution at low ($u = 0.10$) and high ($u = 0.85$) ambiguity, each normalized by mean output at low ambiguity. Three features stand out. First, the entire mass shifts left as tighter credit lowers equilibrium investment. Second, and more important for our purpose, the lower tail thickens disproportionately: the fifth percentile (vertical lines) moves left and, over the full ambiguity range, $Y_{0.05}$ falls by about 14%—a decline that dwarfs the change in the mean and is the Growth-at-Risk content of the model. Third, both densities are visibly bimodal, with a left "default" mode and a right "survival" mode generated by the mixture in (6); raising ambiguity reallocates probability from the survival mode toward the default mode, which is precisely how downside risk is manufactured here.

Figure 3. Output distribution at low vs. high ambiguity



Notes: Blue: $u = 0.10$. Orange: $u = 0.85$. Output is normalized by mean output at low ambiguity. Vertical lines mark $Y_{0.05}$, which shifts left as ambiguity rises (about a 14% decline over the full range). The left and right humps are the default and survival modes of the mixture (6). Based on 120,000 Monte Carlo draws.

5 COUNTERFACTUALS

The counterfactuals switch off one channel at a time so that each figure isolates the contribution of a single mechanism to the lower tail. Throughout, the comparison object is downside risk $Y_{0.05}$, and the diagnostic is how far each shut-down lifts the lower tail relative to the full model.

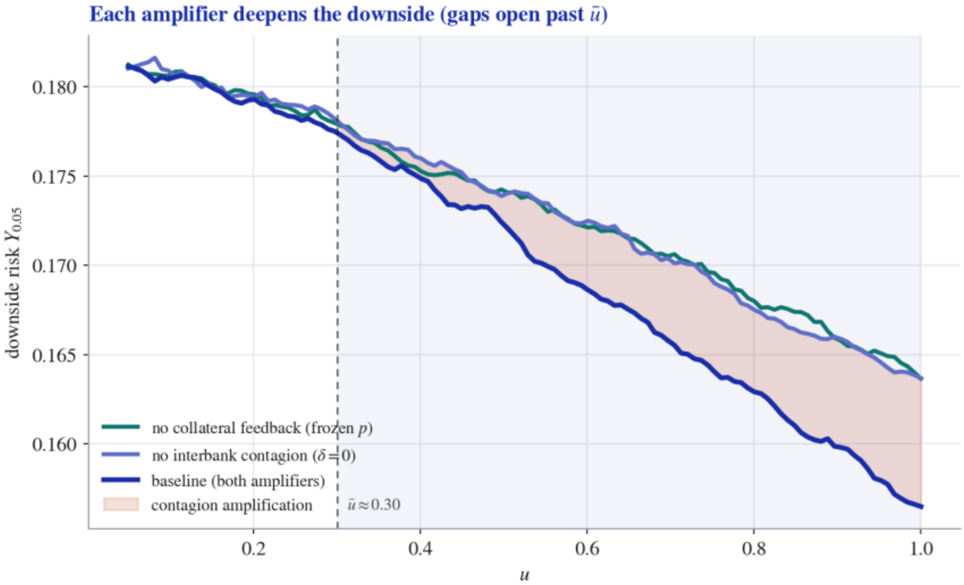
5.1 Shutting down the amplifiers

Figure 4 traces downside risk $Y_{0.05}$ across the whole ambiguity range under the full model and under two channel shut-downs, so that the vertical gap at each u is the contribution of one amplifier. The lowest (baseline) curve has both amplifiers active. Setting $\delta = \ell\beta = 0$ severs the interbank claim while leaving every direct credit response intact: the resulting curve coincides with the baseline below $\bar{u}$ and lifts above it for $u > \bar{u}$, and the shaded wedge between the two is, by construction, the pure contribution of fire-sale contagion to downside risk—it opens precisely at the endogenous solvency threshold and widens thereafter. Freezing the collateral price at its low-ambiguity level (no feedback) lifts the curve similarly: with the price held up, pledgeable wealth does not erode as credit tightens, so the additional collateral-driven contraction is removed. That both shut-downs raise $Y_{0.05}$ confirms that each channel deepens the tail, and that the two are quantitatively modest individually but reinforcing: consistent with the joint amplification emphasized in Section 2. The picture also makes plain why the channels belong in one model: the contagion wedge is invisible until $\bar{u}$, exactly where the collateral feedback is already biting.

5.2 Housing fundamentals and the collateral feedback

Because the collateral price is now an equilibrium object (Lemma 3), we cannot vary it directly; instead we vary a fundamental, the housing-supply elasticity, and let market clearing translate it into a price path. Figure 5 compares a tight (inelastic, $\varepsilon = 0.6$) and an ample (elastic, $\varepsilon = 3.0$) housing market, with the two supply schedules anchored to the same baseline price at low ambiguity so they differ only in how the price responds. Panel (a) plots the equilibrium price $p^*(u)$: both start together, but the inelastic market falls roughly twice as far (about -4% versus

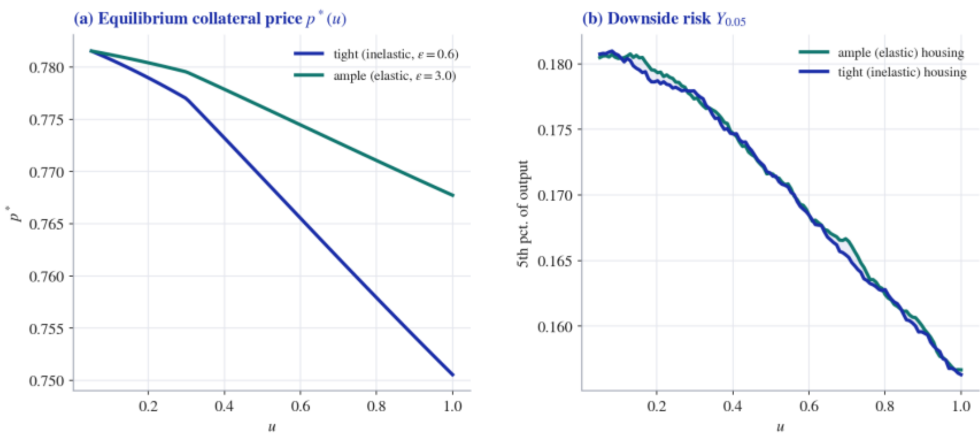
Figure 4. Downside risk under counterfactual channel shut-downs



Notes: $Y_{0.05}$ against ambiguity u. Baseline (both amplifiers active) lies lowest. "No interbank contagion" sets $\delta = 0$; "no collateral feedback" freezes the collateral price at its low-ambiguity level. Each shut-down raises $Y_{0.05}$, and the gaps open beyond the endogenous threshold $\bar{u} \approx 0.30$ (shaded: contagion amplification).

-2%) and visibly bends downward at $\bar{u} \approx 0.30$, where contagion accelerates the credit contraction that feeds back into housing demand. Panel (b) plots the resulting downside risk: the inelastic economy carries a consistently lower $Y_{0.05}$, and the gap is widest in the intermediate-to-high ambiguity region where the feedback multiplier $1/(b_h - \gamma_f \mathcal{I}_p)$ is largest. The widening gap is the graphical content of the cross-partial in Proposition 4: weak collateral fundamentals and high ambiguity are complements, not merely additive. The downside effect is quantitatively modest, the collateral channel operates primarily through the price in panel (a), but it is consistent in sign and direction. We stress that "tight" and "ample" refer to equilibrium prices generated by different fundamentals, so no comparative static treats the price as exogenous.

Figure 5. Collateral feedback: downside risk under tight vs. ample housing fundamentals

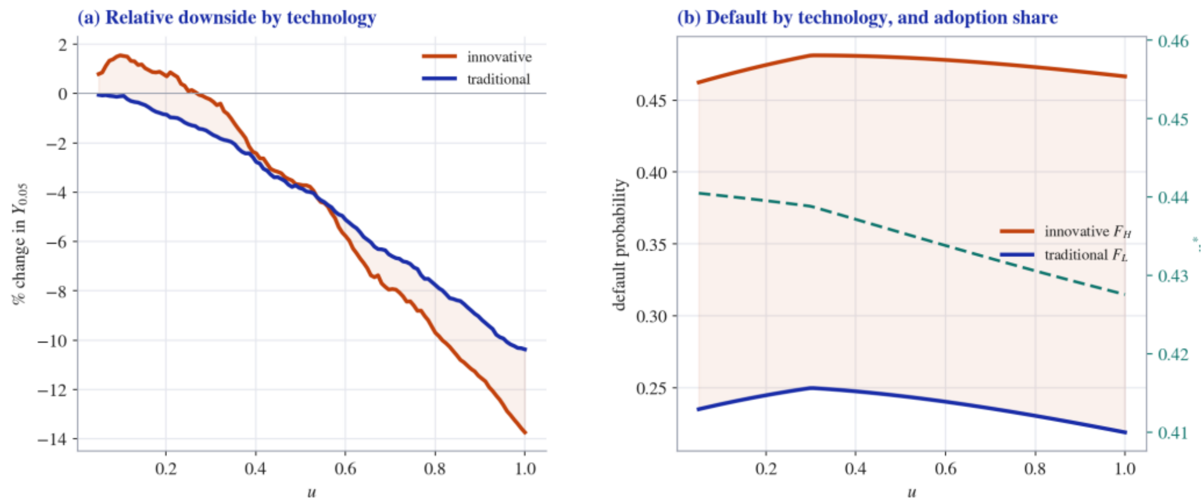


Notes: Tight (inelastic, $\varepsilon = 0.6$) vs. ample (elastic, $\varepsilon = 3.0$) housing supply, anchored to a common baseline price. Panel (a): equilibrium collateral price $p^*(u)$, the inelastic market falls about twice as far and bends at $\bar{u}$. Panel (b): downside risk $Y_{0.05}$, inelastic housing carries a lower tail, with the gap widest at intermediate-to-high ambiguity (Proposition 4).

5.3 Who bears the tail risk

Figure 6 disaggregates downside risk by technology. Panel (a) plots the relative change in $Y_{0.05}$ for each type as ambiguity rises: the innovative line falls away faster, so that from low to high ambiguity the innovative downside deteriorates by roughly 14% against about 10% for the traditional sector, and the gap widens with u, the visual statement of Proposition 5 and Corollary 5. The gap is driven jointly by the three technology primitives, lower recovery r^H_{rec} (which raises the actuarial part of the innovative spread), higher frontier risk σ_H (the ambiguity part), and higher capital intensity α_H (which converts a given credit loss into a larger output loss). Panel (b) plots default probabilities by type together with the equilibrium adoption share κ^* (right axis): innovative firms default far more often throughout, $F_H \approx 0.47$ against $F_L \approx 0.23$, and the adoption share drifts down as ambiguity rises (from about 0.44 to 0.43), the technology-choice margin of Proposition 5. The decline in κ^* is modest, so the heterogeneity result rests mainly on the persistent vulnerability gap in default and downside rather than on a large compositional shift; the contrast between the panels shows that the innovative sector's extra fragility operates through the severity of tail outcomes, not only the frequency of default.

Figure 6. Heterogeneous firms and ambiguity

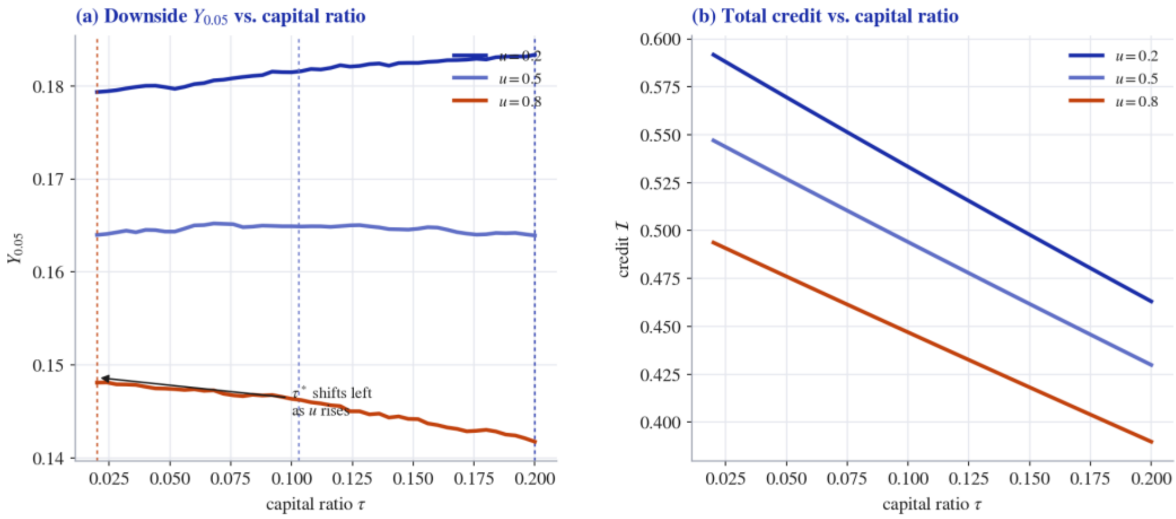


Notes: Panel (a): relative change in $Y_{0.05}$ by technology—the innovative curve falls faster and the gap widens with u (Corollary 5). Panel (b): default probability by technology (left axis) and the equilibrium adoption share κ^* (right axis, dashed)—innovative firms default far more, and κ^* drifts down modestly as ambiguity rises. Innovative: $(\alpha_H, \sigma_H, r^H_{\text{rec}}) = (0.55, 0.6, 0.3)$; traditional: $(0.45, 0.4, 0.6)$.

5.4 State-contingent capital policy

Figure 7 evaluates the capital requirement across ambiguity regimes and is the quantitative counterpart of Proposition 6. Panel (a) plots $Y_{0.05}$ against τ for $u \in \{0.2, 0.5, 0.8\}$. The trade-off reverses sign across regimes, so the optimal buffer shifts sharply left as ambiguity rises: at low ambiguity downside risk is increasing in τ over the whole range (the stabilization benefit dominates, $\tau^* \approx 0.20$); at moderate ambiguity the curve is hump-shaped with an interior optimum near $\tau^* \approx 0.10$; and at high ambiguity it is decreasing throughout (the credit cost dominates, $\tau^* \approx 0.02$). The ordering $\tau^*(0.8) < \tau^*(0.5) < \tau^*(0.2)$ is the figure's main message and the content of $d\tau^*/du < 0$: when ambiguity is high, the dominant drivers of the tail are the ambiguity pass-through and contagion, which extra capital does not offset, so the optimal buffer is lower. Panel (b) plots total credit against τ and explains why: credit is decreasing in τ at every ambiguity level (the cost of capital), and because credit is already scarce when ambiguity is high, the same buffer destroys a larger share of lending in stress, so its marginal credit cost bites hardest exactly when its marginal stabilization benefit is smallest. Together the panels make the case for countercyclical capital: build buffers in calm periods (high τ^*, low credit cost) and release them in stressed periods (low τ^*, where tightening mostly contracts credit without removing the tail).

Figure 7. Macroprudential effectiveness across ambiguity regimes



Notes: Panel (a): $Y_{0.05}$ vs. the capital ratio τ at $u \in \{0.2, 0.5, 0.8\}$; the optimum τ^* (dashed verticals) shifts left as u rises, $\tau^* \approx 0.20, 0.10, 0.02$, so $d\tau^*/du < 0$. Panel (b): total credit vs. τ, decreasing at every ambiguity level; the proportional credit cost of a buffer is largest in stress.

6 CONCLUSION

This paper develops a theoretical framework in which financial uncertainty affects the macroeconomy not only by reducing average credit, but by reshaping the lower tail of output. The central mechanism is ambiguity. When intermediaries become unsure about the probability law governing firm fundamentals, they behave as if their effective risk aversion has increased. This raises credit spreads, reduces lending, and pushes firms closer to default. Because shadow banks rely more heavily on rollover-sensitive funding, their exposure to ambiguity is stronger than that of regulated commercial banks. Shadow credit therefore contracts more sharply when uncertainty rises. Once the shadow bank's net worth is exhausted, interbank claims are marked down through fire-sale losses, transmitting the shock back to the regulated banking sector. At the same time, declining credit demand depresses the equilibrium collateral price, weakens pledgeable wealth, and generates an additional credit–collateral feedback loop. These channels jointly explain why financial uncertainty is most visible in the left tail of aggregate outcomes.

The paper makes four main points. First, financial uncertainty can be given a decision-theoretic foundation: ambiguity is not simply an exogenous spread shock, but a change in lenders' model confidence that endogenously increases effective risk aversion. Second, shadow banking amplifies uncertainty because funding fragility makes shadow intermediaries more sensitive to ambiguous survival probabilities. Third, collateral prices and contagion thresholds are equilibrium objects rather than exogenous amplifiers. The model therefore shows how the same ambiguity shock can pass through credit spreads, shadow-bank balance sheets, interbank exposures, and collateral values before appearing as a deterioration in downside output. Fourth, the real effects of financial uncertainty are heterogeneous across firms. Innovative firms are hit harder because their technologies are more capital intensive, riskier at the frontier, and less pledgeable in default. Financial uncertainty therefore damages not only the level of current output, but also the economy's allocation toward innovation-intensive production.

The numerical results reinforce these theoretical mechanisms. As ambiguity rises, credit spreads increase, credit supply falls, and the effective risk aversion of shadow banks rises more steeply than that of commercial banks. The output distribution shifts left, and the fifth percentile of output declines substantially, confirming that the main macroeconomic cost of uncertainty is a left-tail cost. Counterfactual exercises show that removing interbank contagion or freezing collateral prices raises the lower tail relative to the full model, indicating that both channels deepen downside risk. The heterogeneous-firm exercises further show that the innovative sector experiences larger tail losses than the traditional sector, while the policy experiments show that the capital ratio that maximizes downside output falls as ambiguity increases.

These results carry several policy implications. First, macroprudential regulation should be state contingent. In low-ambiguity states, higher capital buffers are valuable because they build resilience before stress arrives. In high-ambiguity states, however, additional capital tightening can become counterproductive: it restricts credit exactly when credit is already scarce, while doing little to offset the ambiguity pass-through and shadow-bank contagion that drive the left tail. This supports a genuinely countercyclical use of capital buffers: build them in calm periods and release them when uncertainty becomes severe.

Second, regulators should monitor shadow banking as a core part of macroprudential policy, not as a peripheral sector outside the banking system. The model shows that shadow-bank fragility matters because rollover-sensitive funding magnifies ambiguity and because interbank claims transmit shadow-bank losses back to regulated banks. A narrow regulatory focus on commercial-bank capital can therefore miss an important source of systemic risk. Stress tests should incorporate shadow-bank funding fragility, interbank exposure, and fire-sale price impact, especially when uncertainty about firm fundamentals is elevated.

Third, collateral-based regulation should recognize the endogeneity of collateral prices. Loan-to-value limits and collateral haircuts may reduce leverage in normal times, but they can also amplify downside risk when falling collateral prices tighten borrowing constraints and depress credit further. The relevant policy question is therefore not whether collateral constraints are useful in general, but when they should be tightened or relaxed. A state-contingent collateral policy should lean against excessive leverage in booms while avoiding excessive procyclical tightening during ambiguity-driven stress.

Fourth, policy should pay special attention to the financing of innovative firms during periods of financial uncertainty. Because innovative firms have lower recovery values and higher frontier risk, they face a larger increase in borrowing costs and a sharper deterioration in downside outcomes. If left unaddressed, ambiguity shocks can generate persistent allocative damage by pushing the economy away from innovation-intensive production. Targeted credit guarantees, public co-financing schemes, or temporary liquidity facilities for high-productivity but low-pledgeability firms may help prevent financial stress from becoming a long-run drag on productivity growth.

Finally, the paper suggests that macroprudential policy should be evaluated using tail-risk criteria rather than mean outcomes alone. A policy that leaves average credit or average output largely unchanged may still have large effects on the lower tail. The fifth percentile of output, or related Growth-at-Risk measures, therefore provides a natural objective for regulators concerned with systemic fragility. Designing policy for the tail is particularly important in economies where financial intermediation is split between regulated banks and shadow banks, and where collateral values move endogenously with credit conditions.

The model is intentionally stylized, and several extensions remain for future research. A dynamic version could allow ambiguity, collateral prices, and intermediary net worth to evolve over time, making it possible to study persistence and recovery after uncertainty shocks. An empirical extension could estimate the model using bank–firm matched data, shadow-bank balance-sheet information, and regional collateral-price variation. Finally, the framework could be extended to study monetary policy interactions, since interest-rate policy may affect both collateral values and shadow-bank funding conditions. These extensions would further clarify how uncertainty travels through modern financial systems. The core lesson, however, is already clear: when the financial system is governed by ambiguity, the central object of policy should not be the average path of credit, but the fragility of the left tail.

REFERENCES

Daron Acemoglu, Asuman Ozdaglar, and Alireza Tahbaz-Salehi. Systemic risk and stability in financial networks. *American Economic Review*, 105(2):564–608, 2015.

Viral V. Acharya, Sreedhar T. Bharath, and Anand Srinivasan. Does industry-wide distress affect defaulted firms? evidence from creditor recoveries. *Journal of Financial Economics*, 85 (3):787–821, 2007.

Tobias Adrian and Adam B. Ashcraft. Shadow banking regulation. *Annual Review of Financial Economics*, 4:99–140, 2012.

Tobias Adrian and Hyun Song Shin. Liquidity and leverage. *Journal of Financial Intermediation*, 19(3):418–437, 2010.

Tobias Adrian, Nina Boyarchenko, and Domenico Giannone. Vulnerable growth. *American Economic Review*, 109(4):1263–1289, 2019.

Shekhar Aiyar, Charles W. Calomiris, and Tomasz Wieladek. Does macro-prudential regulation leak? evidence from a uk policy experiment. *Journal of Money, Credit and Banking*, 46(s1): 181–214, 2014.

Ozge Akinci and Jane Olmstead-Rumsey. How effective are macroprudential policies? an empirical investigation. *Journal of Financial Intermediation*, 33:33–57, 2018.

Yunus Aksoy and Henrique S. Basso. Liquidity, term spreads and monetary policy. *The Economic Journal*, 124(581):1234–1278, 2014.

Franklin Allen and Douglas Gale. Financial contagion. *Journal of Political Economy*, 108(1): 1–33, 2000.

Heitor Almeida and Murillo Campello. Financial constraints, asset tangibility, and corporate investment. *Review of Financial Studies*, 20(5):1429–1460, 2007.

Yener Altunbas, Mahir Binici, and Leonardo Gambacorta. Macroprudential policy and bank risk. *Journal of International Money and Finance*, 81:203–220, 2018.

Scott R. Baker, Nicholas Bloom, and Steven J. Davis. Measuring economic policy uncertainty. *The Quarterly Journal of Economics*, 131(4):1593–1636, 2016.

Susanto Basu and Brent Bundick. Uncertainty shocks in a model of effective demand. *Econometrica*, 85(3):937–958, 2017.

Ben S. Bernanke, Mark Gertler, and Simon Gilchrist. The financial accelerator in a quantitative business cycle framework. In John B. Taylor and Michael Woodford, editors, *Handbook of Macroeconomics*, volume 1, Part C, pages 1341–1393. Elsevier, 1999.

Javier Bianchi and Enrique G. Mendoza. Optimal time-consistent macroprudential policy. *Journal of Political Economy*, 126(2):588–634, 2018.

Nicholas Bloom. The impact of uncertainty shocks. *Econometrica*, 77(3):623–685, 2009.

Nicholas Bloom. Fluctuations in uncertainty. *Journal of Economic Perspectives*, 28(2):153–176, 2014.

Claudio Borio and Haibin Zhu. Capital regulation, risk-taking and monetary policy: A missing link in the transmission mechanism? *Journal of Financial Stability*, 8(4):236–251, 2012.

James R. Brown, Steven M. Fazzari, and Bruce C. Petersen. Financing innovation and growth: Cash flow, external equity, and the 1990s r&d boom. *The Journal of Finance*, 64(1):151–185, 2009.

Valentina Bruno and Hyun Song Shin. Cross-border banking and global liquidity. *The Review of Economic Studies*, 82(2):535–564, 2015.

Dario Caldara, Cristina Fuentes-Albero, Simon Gilchrist, and Egon Zakrajšek. The macroeconomic impact of financial and uncertainty shocks. *European Economic Review*, 88:185–207, 2016.

Robert E. Carpenter and Bruce C. Petersen. Capital market imperfections, high-tech investment, and new equity financing. *The Economic Journal*, 112(477):F54–F72, 2002.

Danilo Cascaldi-Garcia, Cisil Sarisoy, Juan M. Londono, Bo Sun, Deepa D. Datta, Thiago Ferreira, Olesya Grishchenko, Mohammad R. Jahan-Parvar, Francesca Loria, Sai Ma, Marius Rodriguez, Ilknur Zer, and John Rogers. What is certain about uncertainty? *Journal of Economic Literature*, 61(2):624–654, 2023.

Eugenio Cerutti, Stijn Claessens, and Luc Laeven. The use and effectiveness of macroprudential policies: New evidence. *Journal of Financial Stability*, 28:203–224, 2017.

Thomas Chaney, David Sraer, and David Thesmar. The collateral channel: How real estate shocks affect corporate investment. *American Economic Review*, 102(6):2381–2409, 2012.

Morris A. Davis and Jonathan Heathcote. Housing and the business cycle. *International Economic Review*, 46(3):751–784, 2005.

Giovanni Dell'Ariccia, Luc Laeven, and Robert Marquez. Real interest rates, leverage, and bank risk-taking. *Journal of Economic Theory*, 149:65–99, 2014.

Larry G. Epstein and Martin Schneider. Ambiguity and asset markets. *Annual Review of Financial Economics*, 2:315–346, 2010.

Jie Gan. Collateral, debt capacity, and corporate investment: Evidence from a natural experiment. *Journal of Financial Economics*, 85(3):709–734, 2007.

Gary Gorton and Andrew Metrick. Securitized banking and the run on repo. *Journal of Financial Economics*, 104(3):425–451, 2012.

Bronwyn H. Hall and Josh Lerner. The financing of r&d and innovation. *Handbook of the Economics of Innovation*, 1:609–639, 2010.

Lars Peter Hansen and Thomas J. Sargent. Robustness. *Princeton University Press*, 2008.

Charles P. Himmelberg and Bruce C. Petersen. R&d and internal finance: A panel study of small firms in high-tech industries. *The Review of Economics and Statistics*, 76(1):38–51, 1994.

Matteo Iacoviello. House prices, borrowing constraints, and monetary policy in the business cycle. *American Economic Review*, 95(3):739–764, 2005.

Matteo Iacoviello and Stefano Neri. Housing market spillovers: Evidence from an estimated dsge model. *American Economic Journal: Macroeconomics*, 2(2):125–164, 2010.

Gabriel Jiménez, Steven Ongena, José-Luis Peydró, and Jesús Saurina. Credit supply and monetary policy: Identifying the bank balance-sheet channel with loan applications. *American Economic Review*, 102(5):2301–2326, 2012.

Gabriel Jiménez, Steven Ongena, José-Luis Peydró, and Jesús Saurina. Hazardous times for monetary policy: What do twenty-three million bank loans say about the effects of monetary policy on credit risk-taking? *Econometrica*, 82(2):463–505, 2014.

Gabriel Jiménez, Steven Ongena, José-Luis Peydró, and Jesús Saurina. Macroprudential policy, countercyclical bank capital buffers, and credit supply: Evidence from the spanish dynamic provisioning experiments. *Journal of Political Economy*, 125(6):2126–2177, 2017.

Kyle Jurado, Sydney C. Ludvigson, and Serena Ng. Measuring uncertainty. *American Economic Review*, 105(3):1177–1216, 2015.

Anil K. Kashyap and Jeremy C. Stein. What do a million observations on banks say about the transmission of monetary policy? *American Economic Review*, 90(3):407–428, 2000.

Anil K. Kashyap, Jeremy C. Stein, and David W. Wilcox. Monetary policy and credit conditions: Evidence from the composition of external finance. *American Economic Review*, 83(1):78–98, 1993.

Amir Kermani and Yueran Ma. Asset specificity of nonfinancial firms. *Quarterly Journal of Economics*, 138(1):205–264, 2023.

Asim Ijaz Khwaja and Atif Mian. Tracing the impact of bank liquidity shocks: Evidence from an emerging market. *American Economic Review*, 98(4):1413–1442, 2008.

Nobuhiro Kiyotaki and John Moore. Credit cycles. *Journal of Political Economy*, 105(2):211–248, 1997.

Peter Klibanoff, Massimo Marinacci, and Sujoy Mukerji. A smooth model of decision making under ambiguity. *Econometrica*, 73(6):1849–1892, 2005.

Zheng Liu, Pengfei Wang, and Tao Zha. Land-price dynamics and macroeconomic fluctuations. *Econometrica*, 81(3):1147–1184, 2013.

Guido Lorenzoni. Inefficient credit booms. *The Review of Economic Studies*, 75(3):809–833, 2008.

Sydney C. Ludvigson, Sai Ma, and Serena Ng. Uncertainty and business cycles: Exogenous impulse or endogenous response? *American Economic Journal: Macroeconomics*, 13(4): 369–410, 2021.

Alan Moreira and Alexi Savov. The macroeconomics of shadow banking. *The Journal of Finance*, 72(6):2381–2432, 2017.

Zoltan Pozsar, Tobias Adrian, Adam Ashcraft, and Hayley Boesky. Shadow banking. Staff Report 458, Federal Reserve Bank of New York, 2010.

Albert Saiz. The geographic determinants of housing supply. *Quarterly Journal of Economics*, 125(3):1253–1296, 2010.

Andrei Shleifer and Robert W. Vishny. Liquidation values and debt capacity: A market equilibrium approach. *Journal of Finance*, 47(4):1343–1366, 1992.

Jing Wu, Joseph Gyourko, and Yongheng Deng. Real estate collateral value and investment: The case of china. *Journal of Urban Economics*, 86:43–53, 2015.

A PROOFS

This appendix gives formal proofs for the lemmas, propositions, and corollaries stated in the main text. Because several results are local comparative-static statements, the proofs are written on a compact ambiguity domain $\mathcal{U} \subset \mathbb{R}_+$ on which all equilibrium objects are interior and differentiable. The following regularity conditions make explicit the restrictions that are used below.

Assumption 2 (Regularity conditions for the appendix)**.** *For each technology $j \in \{L, H\}$ and intermediary $k \in \{B, S\}$:*

(i) Terminal wealth in all repayment states is strictly positive; F_j has a strictly positive C^2 density f_j on the relevant support; and all policy choices considered below are interior.

(ii) Let

$$a_k(u) \equiv \theta_k(u)u.$$

The ambiguity wedge satisfies $a_k'(u) > 0$ and $a_k''(u) \geq 0$ on $\mathcal{U}$. Moreover, the shadow-bank pass-through is strong enough that its credit elasticity exceeds the commercial-bank elasticity:

$$\frac{u\zeta a_S'(u)}{1+\zeta a_S(u)} > \frac{u\xi a_B'(u)}{\tau+\xi a_B(u)} - \frac{uA_B'(u)}{A_B(u)}, \qquad A_B(u) \equiv \tilde{N}_B(u)(1+\tau) + \lambda p^*(u)\chi.$$

(iii) The local curvature of the credit fixed point is such that

$$\mathcal{I}_u(u) < 0, \qquad \mathcal{I}_{uu}(u) < 0.$$

A sufficient primitive condition is that the curvature generated by the ambiguity wedge and by the collateral-price feedback dominates the convexity of the reciprocal supply schedules:

$$\sum_{k\in\{B,S\}} \left[A_k D_k D_k'' - A_k'' D_k^2 + 2A_k' D_k D_k' - 2A_k (D_k')^2\right] > 0,$$

where $b_k = A_k/D_k$, $D_B = \tau + \xi a_B$, $D_S = 1 + \zeta a_S$, $A_S = N_S$, and derivatives are with respect to u.

(iv) The spread channel dominates the scale channel in the default threshold:

$$\frac{d\log(R + \Delta R_k^j + M)}{du} > (1-\alpha_j)\left|\frac{d\log I_j^*}{du}\right|,$$

and the threshold is locally convex, $d^2\bar{\omega}_j^*/du^2 \geq 0$.

(v) For $u > \bar{u}$, the interbank write-down is increasing:

$$b_S^*(u) + (u-\bar{u})\frac{db_S^*(u)}{du} > 0.$$

(vi) The output quantile $Y_{0.05}$ is increasing in credit and collateral value and decreasing in default thresholds. Its cross-partials have the signs used in the statements below: collateral strength weakens the marginal downside effect of ambiguity, and the high-technology sector has larger ambiguity exposure because of its lower recovery rate, higher dispersion, and higher capital intensity.

(vii) *The policy objective* $Y_{0.05}(\tau, \lambda \mid u)$ *is twice continuously differentiable, strictly concave in* τ *at the optimum, and has decreasing differences in* (τ, u)*:*

$$\frac{\partial^2 Y_{0.05}}{\partial \tau^2} < 0, \qquad \frac{\partial^2 Y_{0.05}}{\partial \tau\, \partial u} < 0.$$

Remark 2 (Notation for ambiguity)**.** Equation (7) writes the second-order variance as $\mathrm{Var}_\mu(s(m)) = \hat{u}^2 \varsigma^2$. The smooth-ambiguity certainty-equivalent penalty is therefore second order in $\hat{u}$. The linear law $\theta_k(u) = \theta_k^0 + \phi_k u$ is exact when u is interpreted as the variance-level ambiguity index $u \equiv \hat{u}^2$. Equivalently, if one keeps $\hat{u}$ as the primitive ambiguity index, every occurrence of $\phi_k u$ in the effective-risk-aversion law should be replaced by $\phi_k \hat{u}^2$. The main text uses the re-scaled notation u.

A.1 Proof of Lemma 1 (Ambiguity as effective risk aversion)

Proof. *Fix an intermediary* k *and suppress the subscript* k *when no confusion can arise. For a model* m*, define the model-contingent certainty index*

$$C(m; b) \equiv s(m)U(W^{\mathrm{norm}}(b)) + \{1 - s(m)\}U(W^{\mathrm{def}}(b)),$$

where b *is the loan position,* $s(m) = 1 - F_j(\bar{\omega}_j \mid m)$ *is the survival probability under model* m*, and* W^{norm} *and* W^{def} *are the lender's terminal wealth in the normal and default states. Let*

$$\bar{C}(b) \equiv \mathbb{E}_\mu[C(m; b)].$$

The smooth-ambiguity objective is

$$\mathcal{V}(b) = \mathbb{E}_\mu[v(C(m; b))], \qquad v(x) = -\frac{1}{\eta} e^{-\eta x}, \quad \eta > 0.$$

A second-order Taylor expansion around $\bar{C}(b)$ *gives*

$$\mathcal{V}(b) = v(\bar{C}(b)) + \frac{1}{2} v''(\bar{C}(b))\, \mathrm{Var}_\mu(C(m; b)) + o(\mathrm{Var}_\mu(C(m; b)))\,,$$

because $\mathbb{E}_\mu[C(m; b) - \bar{C}(b)] = 0$*. Define the ambiguity-adjusted certainty equivalent* $CE(b)$ *by* $v(CE(b)) = \mathcal{V}(b)$*. Expanding* $v(CE(b))$ *around* $\bar{C}(b)$ *and using*

$$-\frac{v''(x)}{v'(x)} = \eta$$

yields

$$CE(b) = \bar{C}(b) - \frac{\eta}{2}\, \mathrm{Var}_\mu(C(m; b)) + o(\mathrm{Var}_\mu(C(m; b)))\,.$$

Since $C(m; b)$ *is affine in* $s(m)$*,*

$$\mathrm{Var}_\mu(C(m; b)) = \Big[U(W^{\mathrm{norm}}(b)) - U(W^{\mathrm{def}}(b))\Big]^2 \mathrm{Var}_\mu(s(m)).$$

Using the variance-level index u *described in Remark 2,*

$$\mathrm{Var}_\mu(s(m)) = u\, \varsigma^2,$$

and therefore

$$CE(b) = \bar{C}(b) - \frac{\eta}{2} \underbrace{\varsigma^2 \Big[U(W^{\text{norm}}(b)) - U(W^{\text{def}}(b))\Big]^2}_{\equiv \mathcal{E}_k(b)^2} u + o(u).$$

The ambiguity penalty is increasing in η *and in the balance-sheet exposure* $\mathcal{E}_k(b)$*.*

It remains to connect this penalty to an effective-risk-aversion coefficient. Let $\Psi(b, \theta)$ *denote the first-order condition of the expected-utility problem with CRRA coefficient* θ*, and let* $\Psi^A(b, u)$ *denote the first-order condition generated by the ambiguity-adjusted certainty equivalent above. At* $u = 0$*, the two problems coincide, so*

$$\Psi^A(b, 0) = \Psi(b, \theta_k^0).$$

Suppose the local identification condition $\Psi_\theta(b, \theta_k^0) \neq 0$ *holds. By the Implicit Function Theorem, there is a differentiable function* $\theta_k(u)$ *satisfying*

$$\Psi(b, \theta_k(u)) = \Psi^A(b, u), \qquad \theta_k(0) = \theta_k^0.$$

Differentiating at $u = 0$ *gives*

$$\theta_k'(0) = \frac{\partial \Psi^A(b, u)/\partial u|_{u=0}}{\Psi_\theta(b, \theta_k^0)} \equiv \phi_k.$$

The ambiguity penalty puts additional weight on the low-wealth default state; under the usual monotone-likelihood-ratio ordering of repayment states this implies $\phi_k > 0$*. Moreover* $\partial\phi_k/\partial\eta > 0$ *and* $\partial\phi_k/\partial\mathcal{E}_k > 0$ *because* $\partial\Psi^A/\partial u$ *is proportional to* $\eta\mathcal{E}_k\mathcal{E}_{k,b}$*. Hence, locally,*

$$\theta_k(u) = \theta_k^0 + \phi_k u + o(u), \qquad \phi_k = \Lambda(\eta, \mathcal{E}_k) > 0,$$

which is the claimed effective-risk-aversion representation. □

A.2 Proof of Lemma 2 (Funding fragility steepens pass-through)

Proof. *From the proof of Lemma 1, the pass-through slope* ϕ_k *is increasing in the exposure term*

$$\mathcal{E}_k = \varsigma \Big[U(W_k^{\text{norm}}) - U(W_k^{\text{def}})\Big].$$

Thus it is sufficient to show $\mathcal{E}_S > \mathcal{E}_B$*.*

Let the commercial bank be funded by stable insured deposits, so that default losses are limited to asset-side losses. Its default-state continuation value can be written as:

$$W_B^{\text{def}} = N_B - (R - r_{\text{rec}}^j) b_B.$$

The shadow bank uses rollover-sensitive wholesale funding. When the pessimistic model has a low survival probability, wholesale creditors refuse to roll over, forcing liquidation before asset payoffs are realized. Let the resulting liquidation loss be $\mathcal{L}(\rho, u) > 0$*, where* $\rho > 0$ *is the wholesale funding share. Conditional on the same loan portfolio, the shadow bank's default-state continuation value is*

$$W_S^{\text{def}} = W_B^{\text{def}} - \mathcal{L}(\rho, u) < W_B^{\text{def}}.$$

For the relevant comparison, hold the survival-state payoff fixed or, more generally, assume that *any survival-state advantage of the shadow bank is smaller than the rollover loss in the default state. Since U is strictly increasing,*

$$U(W_S^{\text{def}}) < U(W_B^{\text{def}}).$$

Therefore

$$U(W_S^{\text{norm}}) - U(W_S^{\text{def}}) > U(W_B^{\text{norm}}) - U(W_B^{\text{def}}),$$

which implies $\mathcal{E}_S > \mathcal{E}_B$. Since $\phi_k = \Lambda(\eta, \mathcal{E}_k)$ is strictly increasing in $\mathcal{E}_k$, it follows that

$$\phi_S > \phi_B.$$

Therefore the steeper ambiguity pass-through of shadow banks is a consequence of their funding structure. □

A.3 Derivation of the intermediary supply schedules

Proof. *For the commercial bank, define*

$$W_B^{\text{def}} = N_B - (R - r_{\text{rec}}^j)b_B, \qquad W_B^{\text{norm}} = N_B + b_B\Delta R_B + \lambda p\chi.$$

The effective expected-utility problem is

$$\max_{b_B} \; F_j(\bar{\omega}_j)\frac{(W_B^{\text{def}})^{1-\theta_B}}{1-\theta_B} + [1 - F_j(\bar{\omega}_j)]\frac{(W_B^{\text{norm}})^{1-\theta_B}}{1-\theta_B}.$$

The first-order condition is

$$-F_j(\bar{\omega}_j)(R - r_{\text{rec}}^j)(W_B^{\text{def}})^{-\theta_B} + [1 - F_j(\bar{\omega}_j)]\Delta R_B(W_B^{\text{norm}})^{-\theta_B} = 0.$$

Hence

$$\Delta R_B = \frac{F_j(\bar{\omega}_j)}{1 - F_j(\bar{\omega}_j)}(R - r_{\text{rec}}^j)\left(\frac{W_B^{\text{norm}}}{W_B^{\text{def}}}\right)^{\theta_B}.$$

This equation is the exact marginal pricing condition. The closed-form supply schedule used in the main text is the local reduced-form representation of this pricing condition combined with the binding capital requirement and the collateral constraint. Writing the local curvature term as $\xi > 0$ gives:

$$b_B^* = \frac{\widetilde{N}_B(1+\tau) + \lambda p\chi}{\tau + \xi\theta_B(u)u}.$$

When contagion is inactive, $\widetilde{N}_B = N_B$. The shadow bank has no capital requirement and no collateral facility in the benchmark. The identical localization of its first-order condition gives:

$$b_S^* = \frac{N_S}{1 + \zeta\theta_S(u)u}, \qquad \zeta > 0.$$

□

A.4 Proof of Lemma 3 (Endogenous collateral price and feedback)

Proof. *The housing market-clearing condition is*

$$H^h(p) + H^f(p, u) = \bar{H}.$$

Using $H^h(p) = a - b_h p$ and $H^f(p,u) = \gamma_f \mathcal{I}(p,u)$, define

$$\Gamma(p,u) \equiv a - b_h p + \gamma_f \mathcal{I}(p,u) - \bar{H}.$$

An equilibrium collateral price satisfies $\Gamma(p^*(u), u) = 0$. Since

$$\Gamma_p(p,u) = -b_h + \gamma_f \mathcal{I}_p(p,u),$$

the stability condition $b_h - \gamma_f \mathcal{I}_p > 0$ implies $\Gamma_p \neq 0$. The Implicit Function Theorem therefore gives a differentiable price function $p^*(u)$ with

$$\frac{dp^*}{du} = -\frac{\Gamma_u}{\Gamma_p} = \frac{\gamma_f \mathcal{I}_u}{b_h - \gamma_f \mathcal{I}_p}.$$

Since $\mathcal{I}_u < 0$ and $b_h - \gamma_f \mathcal{I}_p > 0$, we obtain

$$\frac{dp^*}{du} < 0.$$

If $\mathcal{I}(p,u)$ is locally affine in p,

$$\mathcal{I}(p,u) = \mathcal{I}(0,u) + \mathcal{I}_p p,$$

then market clearing can be solved explicitly:

$$a - b_h p + \gamma_f \{\mathcal{I}(0,u) + \mathcal{I}_p p\} = \bar{H},$$

so

$$p^*(u) = \frac{a + \gamma_f \mathcal{I}(0,u) - \bar{H}}{b_h - \gamma_f \mathcal{I}_p}.$$

The multiplier

$$\frac{1}{b_h - \gamma_f \mathcal{I}_p}$$

is increasing as $\gamma_f \mathcal{I}_p$ approaches b_h from below. Hence the closer the collateral-demand response is to the stabilizing slope of household housing demand, the stronger the feedback from credit to collateral prices and back to credit. □

A.5 Proof of the contagion equation (16)

Proof. *The commercial bank holds an interbank claim on the shadow bank equal to*

$$L_{\text{int}} = \beta b_S.$$

For $u \leq \bar{u}$, shadow-bank net worth is nonnegative, the claim is repaid at par, and the commercial bank's effective capital is $\widetilde{N}_B = N_B$.

For $u > \bar{u}$, shadow-bank net worth is exhausted and the shadow bank must liquidate assets at the fire-sale price

$$q(u) = 1 - \ell(u - \bar{u}).$$

The loss rate on the interbank claim is therefore

$$1 - q(u) = \ell(u - \bar{u}).$$

The commercial bank's write-down is

$$[1 - q(u)]L_{\text{int}} = \ell(u - \bar{u})\beta b_S = \delta(u - \bar{u})b_S, \qquad \delta \equiv \ell\beta.$$

Combining the no-distress and distress regions gives

$$\widetilde{N}_B(u) = N_B - \delta(u - \bar{u})b_S\, \mathbf{1}\{u > \bar{u}\},$$

which is Equation (16). □

A.6 Proof of Proposition 1 and Corollary 1

Proof. *Write the commercial-bank supply schedule as*

$$b_B^*(u) = \frac{A_B(u)}{D_B(u)}, \qquad A_B(u) = \widetilde{N}_B(u)(1+\tau) + \lambda p^*(u)\chi, \qquad D_B(u) = \tau + \xi a_B(u),$$

where $a_B(u) = \theta_B(u)u$*. Under Assumption* 2*,* $D_B(u) > 0$*,* $D_B'(u) = \xi a_B'(u) > 0$*, and*

$$A_B'(u) = (1+\tau)\widetilde{N}_B'(u) + \lambda\chi p_u^*(u) \le 0.$$

The inequality is strict whenever either collateral prices fall with ambiguity or contagion is active. Differentiating,

$$\frac{db_B^*}{du} = \frac{A_B' D_B - A_B D_B'}{D_B^2} < 0.$$

For the shadow bank,

$$b_S^*(u) = \frac{N_S}{D_S(u)}, \qquad D_S(u) = 1 + \zeta a_S(u),$$

and therefore

$$\frac{db_S^*}{du} = -\frac{N_S D_S'}{D_S^2} = -\frac{N_S \zeta a_S'(u)}{D_S^2} < 0.$$

Thus both bank and shadow-bank credit contract with ambiguity.

The elasticity of commercial-bank lending with respect to ambiguity is

$$\varepsilon_B(u) \equiv -u\frac{d\log b_B^*}{du} = u\left[\frac{D_B'}{D_B} - \frac{A_B'}{A_B}\right],$$

while the shadow-bank elasticity is

$$\varepsilon_S(u) \equiv -u\frac{d\log b_S^*}{du} = u\frac{D_S'}{D_S}.$$

Assumption 2*(ii) is precisely the sufficient condition* $\varepsilon_S(u) > \varepsilon_B(u)$*. Hence shadow credit is more uncertainty-elastic.*

For the corollary, total credit to type-j *firms is*

$$I_j^*(u) = b_B^{j*}(u) + b_S^{j*}(u).$$

Since each component is decreasing,

$$\frac{\partial I_j^*}{\partial u} < 0.$$

Differentiating once more gives
$$\frac{\partial^2 I_j^*}{\partial u^2} = \frac{\partial^2 b_B^{j*}}{\partial u^2} + \frac{\partial^2 b_S^{j*}}{\partial u^2}.$$
The curvature restriction in Assumption 2(iii) imposes the economically relevant accelerating-contraction region,
$$\frac{\partial^2 I_j^*}{\partial u^2} < 0.$$
Without this additional curvature condition, the reciprocal supply schedules alone do not guarantee a negative second derivative. □

A.7 Proof of Proposition 2 and Corollary 2

Proof. *Let*
$$S_k^j(u) \equiv R + \Delta R_k^j(u) + M.$$
The default threshold can be written as
$$\bar{\omega}_j^*(u) = \left(I_j^*(u)\right)^{1-\alpha_j} S_k^j(u).$$
Taking logarithms,
$$\log \bar{\omega}_j^* = (1-\alpha_j)\log I_j^* + \log S_k^j.$$
Differentiating with respect to u gives
$$\frac{d\log \bar{\omega}_j^*}{du} = (1-\alpha_j)\frac{d\log I_j^*}{du} + \frac{d\log S_k^j}{du}.$$
The first term is negative because credit contracts with ambiguity. The second term is positive because the spread contains the ambiguity component $\psi_k \theta_k(u)\sigma_j$ and $\theta_k'(u) > 0$. By Assumption 2(iv), the spread channel dominates the scale channel:
$$\frac{d\log S_k^j}{du} > (1-\alpha_j)\left|\frac{d\log I_j^*}{du}\right|.$$
Therefore
$$\frac{d\log \bar{\omega}_j^*}{du} > 0, \qquad \textit{and hence} \qquad \frac{d\bar{\omega}_j^*}{du} > 0.$$
Since F_j is strictly increasing with density $f_j > 0$,
$$\frac{dF_j(\bar{\omega}_j^*)}{du} = f_j(\bar{\omega}_j^*)\frac{d\bar{\omega}_j^*}{du} > 0.$$

For the corollary, differentiate again:
$$\frac{d^2 F_j(\bar{\omega}_j^*)}{du^2} = f_j'(\bar{\omega}_j^*)\left(\frac{d\bar{\omega}_j^*}{du}\right)^2 + f_j(\bar{\omega}_j^*)\frac{d^2\bar{\omega}_j^*}{du^2}.$$
When $\bar{\omega}_j^*$ lies below the mode of f_j, the density is increasing, so $f_j'(\bar{\omega}_j^*) > 0$. By Assumption 2(iv), $d^2\bar{\omega}_j^*/du^2 \geq 0$. Since $f_j > 0$, both terms are nonnegative and the first is strictly positive. Hence
$$\frac{d^2 F_j(\bar{\omega}_j^*)}{du^2} > 0.$$

□

A.8 Proof of Proposition 3 and Corollary 3

Proof. *For $u > \bar{u}$, total credit can be written as*

$$\mathcal{I}^*(u) = b_B^*(u, \widetilde{N}_B(u)) + b_S^*(u).$$

Taking the total derivative,

$$\frac{d\mathcal{I}^*}{du} = \underbrace{\frac{\partial b_B^*}{\partial u}}_{\text{(I)}} + \underbrace{\frac{db_S^*}{du}}_{\text{(II)}} + \underbrace{\frac{\partial b_B^*}{\partial \widetilde{N}_B}\frac{d\widetilde{N}_B}{du}}_{\text{(III)}}.$$

The first two terms are negative by Proposition 1. Moreover,

$$\frac{\partial b_B^*}{\partial \widetilde{N}_B} = \frac{1+\tau}{D_B(u)} > 0.$$

From Equation (16), for $u > \bar{u}$,

$$\frac{d\widetilde{N}_B}{du} = -\delta\left[b_S^*(u) + (u-\bar{u})\frac{db_S^*(u)}{du}\right].$$

By Assumption 2(v), the bracketed term is positive, so $d\widetilde{N}_B/du < 0$. Hence term (III) is strictly negative. Therefore total credit contracts more strongly than in the benchmark $\delta = 0$, where term (III) is absent.

At $u = \bar{u}$, the right derivative of $\widetilde{N}_B$ is

$$\left.\frac{d\widetilde{N}_B}{du}\right|_{\bar{u}^+} = -\delta b_S^*(\bar{u}),$$

whereas the left derivative is zero. Thus the first derivative of total credit has a downward kink at $\bar{u}$. Equivalently, under a smooth approximation to the indicator function in Equation (16), the second derivative contains a negative local component at the contagion threshold.

For the corollary, the absolute magnitude of the contagion term is

$$|\text{(III)}| = \frac{1+\tau}{D_B(u)}\delta\left[b_S^*(u) + (u-\bar{u})b_S^{*\prime}(u)\right].$$

Because $\delta = \ell\beta$, this term is increasing in the fire-sale price impact ℓ and in interbank exposure β. It is also increasing in the shadow-bank market share because the bracketed term is increasing in the level of shadow-bank lending. Therefore

$$\mathcal{M}(u) = 1 + \frac{|\text{(III)}|}{|\text{(I)}| + |\text{(II)}|} > 1$$

for $u > \bar{u}$, and the multiplier is increasing in ℓ, β, and the shadow-bank market share. □

A.9 Proof of Proposition 4 and Corollary 4

Proof. *The sign $dp^*/du < 0$ follows directly from Lemma 3. We now show how the collateral-feedback multiplier affects downside risk.*

Let the downside quantile be written locally as a differentiable reduced-form function

$$Y_{0.05} = Q(\mathcal{I}(u, p^*), p^*(u), \bar{\omega}(u)),$$

where $Q_{\mathcal{I}} > 0$*,* $Q_p > 0$*, and* $Q_{\bar{\omega}} < 0$*. Since ambiguity raises default thresholds and reduces credit and collateral values,*

$$\frac{dY_{0.05}}{du} < 0.$$

The collateral-price channel enters through

$$\frac{dp^*}{du} = \frac{\gamma_f \mathcal{I}_u}{b_h - \gamma_f \mathcal{I}_p}.$$

Thus the collateral component of the ambiguity effect is proportional to

$$\frac{1}{b_h - \gamma_f \mathcal{I}_p}.$$

As $b_h - \gamma_f \mathcal{I}_p$ *becomes smaller, this multiplier becomes larger, so a given ambiguity shock induces a larger fall in* p^**, a larger fall in pledgeable collateral* $\lambda p^* \chi$*, and a larger decline in* $Y_{0.05}$*.*

Assumption 2(vi) states the corresponding complementarity condition in cross-partial form:

$$\frac{\partial^2 Y_{0.05}}{\partial u \, \partial p^*} > 0.$$

Since $Y_u < 0$*, this means a higher collateral price makes the marginal effect of ambiguity less negative; equivalently, weak collateral values magnify downside risk.*

A sign qualification is needed for the housing-supply parameter $\bar{H}$*. From the market-clearing equation,*

$$\frac{\partial p^*}{\partial \bar{H}} = -\frac{1}{b_h - \gamma_f \mathcal{I}_p} < 0.$$

Therefore, if $\bar{H}$ *is literally interpreted as housing supply, then*

$$\frac{\partial^2 Y_{0.05}}{\partial u \, \partial \bar{H}} = \frac{\partial^2 Y_{0.05}}{\partial u \, \partial p^*} \frac{\partial p^*}{\partial \bar{H}} < 0.$$

Thus the statement $\partial^2 Y_{0.05}/(\partial u \, \partial \bar{H}) > 0$ *is valid only if* $\bar{H}$ *is reinterpreted as an inverse measure of collateral scarcity or collateral weakness. With* $\bar{H}$ *as physical housing supply, the correct sign is negative.*

For the LTV corollary, bank lending depends on λ *through*

$$b_B^* = \frac{\widetilde{N}_B(1+\tau) + \lambda p^*(u)\chi}{D_B(u)}.$$

Thus

$$\frac{\partial b_B^*}{\partial \lambda} = \frac{p^*(u)\chi}{D_B(u)} > 0.$$

A higher LTV cap relaxes collateral constraints and makes the marginal ambiguity effect on $Y_{0.05}$ *less negative. Under Assumption 2(vi),*

$$\frac{\partial^2 Y_{0.05}}{\partial u \, \partial \lambda} > 0.$$

Consequently, lowering λ tightens the LTV cap and amplifies the ambiguity–downside-risk transmission. □

A.10 Proof of Proposition 5 and Corollary 5

Proof. *The spread for technology j is*

$$\Delta R_k^j = \frac{F_j(\bar{\omega}_j)(R - r_{\text{rec}}^j)}{1 - F_j(\bar{\omega}_j)} + \psi_k \theta_k(u) \sigma_j.$$

Differentiate with respect to u:

$$\frac{d\Delta R_k^j}{du} = \frac{d}{du}\left[\frac{F_j(\bar{\omega}_j)(R - r_{\text{rec}}^j)}{1 - F_j(\bar{\omega}_j)}\right] + \psi_k \sigma_j \theta_k'(u).$$

The second term is larger for the high technology because $\sigma_H > \sigma_L$. The actuarial term is also larger for the high technology because $r_{\text{rec}}^H < r_{\text{rec}}^L$, so the loss given default $R - r_{\text{rec}}^j$ is larger. Finally, $\alpha_H > \alpha_L$ implies that the high technology is more sensitive to credit contractions through the production term $I_j^{\alpha_j}$ and the financing scale required to operate the technology.

By Assumption 2(vi), these three primitive gaps imply the ordered quantile response

$$\left|\frac{\partial Y_{0.05}^H}{\partial u}\right| > \left|\frac{\partial Y_{0.05}^L}{\partial u}\right|.$$

It remains to prove that the adoption share falls. The adoption share is

$$\kappa^*(u) = G(\Pi_H(u) - \Pi_L(u)).$$

Since G is a distribution function with density $g \geq 0$ on the relevant support,

$$\frac{d\kappa^*}{du} = g(\Pi_H - \Pi_L)\left(\frac{d\Pi_H}{du} - \frac{d\Pi_L}{du}\right).$$

The high technology has lower pledgeability, higher dispersion, and higher capital intensity. These features imply that its profit is more negatively affected by ambiguity:

$$\frac{d\Pi_H}{du} < \frac{d\Pi_L}{du}.$$

Therefore

$$\frac{d\kappa^*}{du} < 0.$$

For the corollary, define the vulnerability gap

$$\mathcal{G}(u) \equiv \left|\frac{\partial Y_{0.05}^H}{\partial u}\right| - \left|\frac{\partial Y_{0.05}^L}{\partial u}\right|.$$

The ordered-spread and ordered-curvature conditions in Assumption 2(vi) imply that the marginal effect of ambiguity steepens faster for the high technology:

$$\frac{d\mathcal{G}(u)}{du} > 0.$$

Hence the vulnerability gap widens with ambiguity. □

A.11 Proof of Proposition 6

Proof. *For a fixed ambiguity level u, define the policy objective*

$$\mathcal{W}(\tau; u) \equiv Y_{0.05}(\tau, \lambda \mid u),$$

with λ held fixed. The capital ratio affects downside risk through two opposing channels. First, a higher capital requirement raises intermediary resilience and reduces the probability that balance-sheet losses propagate. Second, it reduces credit supply. The commercial-bank supply derivative is

$$\frac{\partial b_B^*}{\partial \tau} = \frac{N_B D_B(u) - [N_B(1+\tau) + \lambda p^*(u)\chi]}{D_B(u)^2}, \qquad D_B(u) = \tau + \xi\theta_B(u)u.$$

Thus the credit-cost channel is negative whenever the numerator is negative, which is the empirically relevant region for a binding capital requirement.

Assumption 2(vii) imposes the standard single-peaked policy problem. Hence there exists an interior optimum $\tau^*(u)$ satisfying

$$\mathcal{W}_\tau(\tau^*(u); u) = 0, \qquad \mathcal{W}_{\tau\tau}(\tau^*(u); u) < 0.$$

By the Implicit Function Theorem,

$$\frac{d\tau^*(u)}{du} = -\frac{\mathcal{W}_{\tau u}(\tau^*(u); u)}{\mathcal{W}_{\tau\tau}(\tau^*(u); u)}.$$

The decreasing-differences condition $\mathcal{W}_{\tau u} < 0$ says that the marginal stabilization benefit of a higher capital ratio declines when ambiguity is higher. Since $\mathcal{W}_{\tau\tau} < 0$ at the optimum,

$$\frac{d\tau^*(u)}{du} < 0.$$

Thus the optimal capital ratio is decreasing in ambiguity. □